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Foreword

This Technical Report has been produced by the 3rd Generation Partnership Project (3GPP).

The present document is part of a multi-part TR as described below:

- Part 1: LI for IMS based services. See TR 33.929-1 [6].
- Part 2: LI for IMS based STIR/SHAKEN. See TR 33.929-2 [7].
- Part 3: LI for messaging services (present document).
- Part 4: LI for data in 5G core. See TR 33.929-4 [8].
- Part 5: LI for ID association caching. See TR 33.929-5 [9].
- Part 6: LI for IMS based RCS. See TR 33.929-6 [10].
- Part 7: LI location acquisition capabilities. See TR 33.929-7 [11].
- Part 8: LI for MMS. See TR 33.929-8 [12].
- Part 9: LI for PTC. See TR 33.929-9 [13].

In Part 1, the illustrations of LI for IMS-based services focus on the LI aspects of IMS sessions which include the architecture topologies and the call flows covering basic sessions, redirected sessions, target non-local ID, conferencing, roaming (local break-out and home-routed roaming).

In Part 2, the illustrations of LI for IMS based STIR/SHAKEN focus on various STIR/SHAKEN related LI reporting scenarios.

In Part 3 (present document), the illustrations of LI for messaging services focus on SMS over NAS and SMS over IP.

In Part 4, the illustrations of LI for data in 5G core focus on the LI aspects of data interception focussing on the PDU session related events including the handover scenarios.

In Part 5, the illustrations of LI for ID association caching focus on the capabilities used to provide the temporary identity to permanent identity (and vice-versa) associations known to the CSP network. The illustrations include the architectural concepts and few examples of use-cases encountering various timing scenarios.

In Part 6, the illustrations of LI for IMS based RCS focus on the architecture topologies and the call flows when the RCS service is offered by the CSP or by a Third Party Provider.

In Part 7, the illustrations of LI location acquisition capabilities focus on the conceptual overview and the flow diagrams for location reporting, LALS and location acquisition.

In Part 8, the illustrations of LI for MMS focus on the conceptual overview and the flow diagrams for MMS.

In Part 9, the illustrations of LI for PTC focus on the conceptual overview and the flow diagrams for MCPTT and PoC, the two commonly referred to as Push to Talk over Cellular (PTC).

Introduction

Unlike the previous generation of LI technical specifications, the latest LI technical specifications (TS 33.126 [2], TS 33.127 [3], TS 33.128 [4]) contain only the normative part of the requirements. However, the implementers may need additional information such as architectural diagrams, conceptual scenarios, flow-diagrams and additional background information to better understand the requirements in those specifications. The present document collects the relevant informative annexes from the previous generation of LI technical specifications and then expands them to include the similar level of details as for latest technologies such as 5G.

1 Scope

The present document provides architectural diagrams, conceptual scenarios, flow-diagrams, examples, and other background information which can be useful to implement the LI functions defined in TS 33.126 [2], TS 33.127 [3] and TS 33.128 [4].

The present document covers the illustrations related LI for Messaging Services that include SMS over NAS and SMS over IP.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.126: "Lawful Interception requirements".

[3] 3GPP TS 33.127: "Lawful Interception (LI) architecture and functions".

[4] 3GPP TS 33.128: "Protocol and procedures for Lawful Interception (LI)".

[5] 3GPP TS 24.341: "Support of SMS over IP networks, Stage 3".

[6] 3GPP TR 33.929-1: "Lawful interception implementation guidance; LI for IMS-based services".

[7] 3GPP TR 33.929-2: "Lawful Interception (LI) implementation guidance; LI for IMS based STIR/SHAKEN".

[8] 3GPP TR 33.929-4: "Lawful Interception (LI) implementation guidance; LI for data in 5G core".

[9] 3GPP TR 33.929-5: "Lawful Interception (LI) implementation guidance; LI for ID association caching ".

[10] 3GPP TR 33.929-6: "Lawful Interception (LI) implementation guidance; LI for IMS based RCS".

[11] 3GPP TR 33.929-7: "Lawful Interception (LI) implementation guidance; LI location acquisition capabilities".

[12] 3GPP TR 33.929-8: "Lawful Interception (LI) implementation guidance; LI for MMS".

[13] 3GPP TR 33.929-9: "Lawful Interception (LI) implementation guidance; LI for PTC".

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

For the purposes of the present document, the following symbols apply:

None.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

3GPP	3rd Generation Partnership Program
CSCF	Call Session Control Function
CSP	Communication Service Provider
GPSI	Generic Public Subscription Identifier
HPLMN	Home PLMN
HSS	Home Subscriber Server
I-CSCF	Interrogating CSCF
IBCF	Interrogating Border Control Function
IMS	IP Multimedia Subsystem
IP	Internet Protocol
IP-SM-GW	IP Short Message Gateway
IRI	Intercept Related Information
LALS	Lawful Access Location Services
LBO	Local Break Out
LI	Lawful Interception
LI_X2	Lawful Interception Internal Interface 2
LI_X3	Lawful Interception Internal Interface 3
LMISF	LI Mirror IMS State Function
LMISF-IRI	LMISF for the handling of IRI
MCPTT	Mission Critical Push To Talk
MMS	Multimedia Message Service
MSC	Mobile Switching Center
NAS	Non-Access Stratum
P-CSCF	Proxy Call Session Control Function
PEI	Permanent Equipment Identifier
PLMN	Public Land Mobile Network
PoC	Push to talk over Cellular
POI	Point Of Interception
PTC	Push to Talk over Cellular
RCS	Rich Communication Suite
RTT	Real Time Text
S-CSCF	Serving Call Session Control Function
SHAKEN	Signature-based Handling of Asserted information using toKENs
SIP	Session Initiation Protocol
SM-SC	Short Message Service Center
SMS	Short Message Service
SMS-GMSC	Short Message Service Gateway MSC
SMSF	SMS-Function
STIR	Secure Telephony Identity Revisited

SUPI	Subscriber Permanent Identifier
UE	User Equipment
VPLMN	Visited PLMN
xIRI	LI_X2 Intercept Related Information

4 Illustrations of LI for messaging services

4.1 General

Clause 4 provides a few architectures and flow-diagrams to illustrate the LI for Messaging services. Such Messaging services include Short Messaging Service (SMS), the Multi-media Messaging Service (MMS) and the Rich Communication Suite (RCS). Even though the Real Time Text (RTT) is a text-based communication, the scope of LI for RTT is presumed to be part of LI for voice services.

MMS and RCS are session-based communications whereas SMS is a session independent communication which basically means from an LI perspective the interception and delivery of each SMS message transfer are handled independently without any attempt to correlate (at the LI-level) with the prior reporting.

Redirection of incoming messages can happen with MMS and RCS, but not with SMS.

The UE involved in the messaging service may be roaming, or non-roaming. In the case of roaming, the LI for MMS is supported only in the HPLMN, the LI for SMS over NAS is supported only in the serving PLMN and the LI for SMS over IP is supported in the VPLMN and the HPLMN. For the SMS over IP case, either the home-routed roaming or the LBO based roaming is possible.

4.2 Short Messaging Service (SMS)

4.2.1 General

Clause 4.2 illustrates the LI for SMS over NAS and SMS over IP, depending on the UE capability. A UE capable of receiving SMS over IP may still receive the SMS over NAS depending on its connectivity at the time the message transfer is attempted.

In the illustrations, SMS over NAS refers to the SMS over 5GS NAS (where the SMSF that serves the UE is involved). The other possible network topology considerations (e.g. SMS over 4G NAS) are outside the scope of these illustrations even though the principles of LI are the same as for SMS over 5GS NAS.

For SMS over IP, the IRI-POI in the S-CSCF provides the LI functions as a default method and the IRI-POI in the P-CSCF provides the LI functions as an alternate method (TS 33.127 [3]). In the HPLMN, the IRI-POI in the S-CSCF provides the LI functions as a default method and the IRI-POI in the P-CSCF (with home-routed) or the IBCF (with LBO) provides the LI functions as an alternate method (TS 33.127 [3]). In the VPLMN with LBO, the IRI-POI in the P-CSCF provides the LI functions and with home-routed, the LMISF-IRI provides the LI functions.

4.2.2 Network topologies

4.2.2.1 UEs are not roaming

Figure 4.2.2.1-1 illustrates all the possible permutations of SMS message transfer scenarios when all of the UEs are not roaming. In the illustration, UE-A1, UE-A2 and UE-A3 are served by PLMN-A and UE-B1, UE-B2, and UE-B3 are served by PLMN-B.

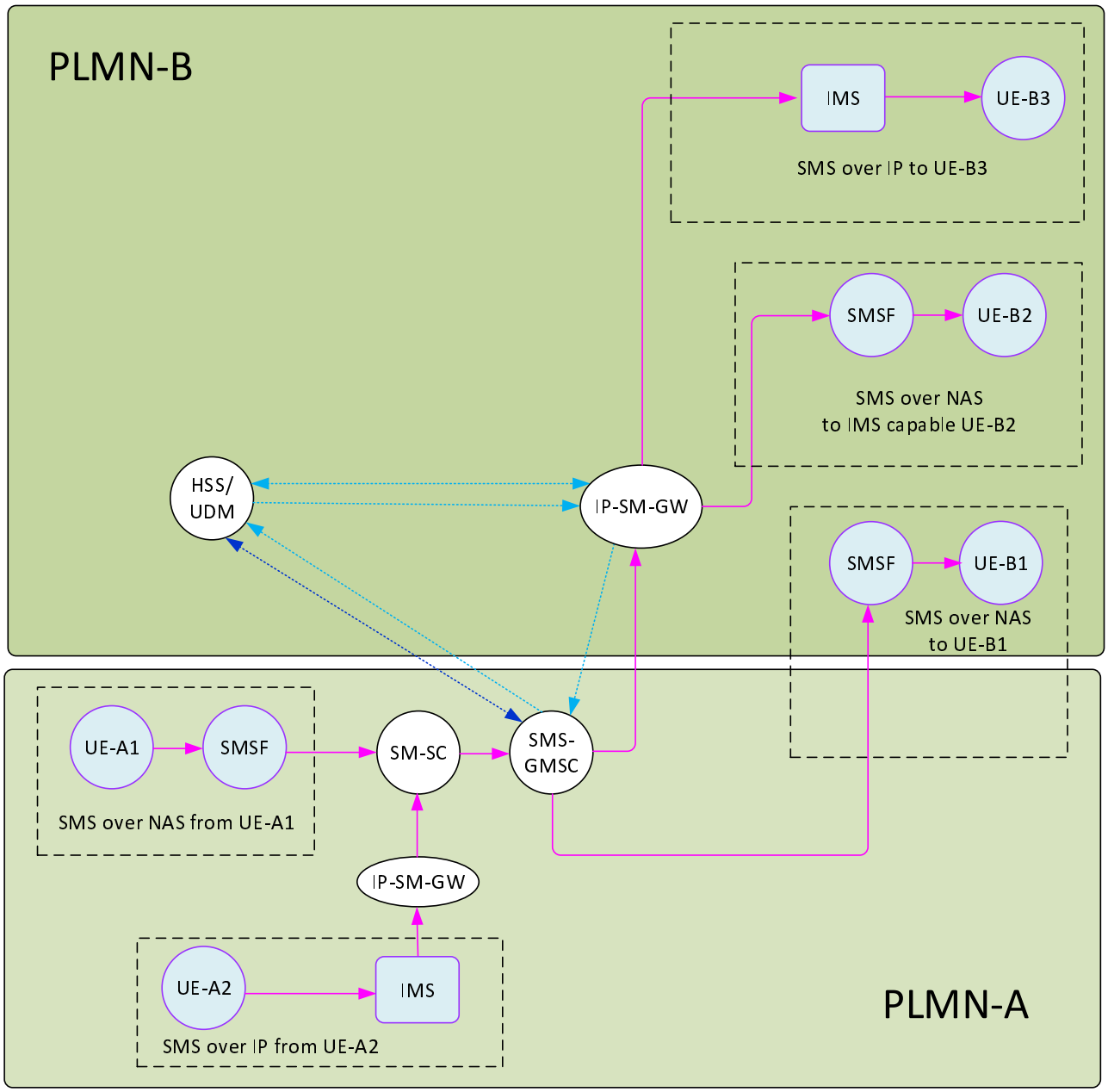


Figure 4.2.2.1-1: SMS message transfer when the UEs are non-roaming

The UE-A1 sends an SMS message over NAS. In this case the SMSF (PLMN-A) that serves the UE-A1 provides the LI functions when UE-A1 is a target in PLMN-A. The SMSF (PLMN-A) that serves the UE-A1 also provides the LI functions when the destination party of the SMS message is a target non-local ID and, in this case, the UE-A1 may not be a target in PLMN-A.

The UE-A2 sends the SMS message over IP. In this case, the IMS signalling function within the IMS domain (PLMN-A) provides the LI functions when UE-A2 is a target in PLMN-A. The IMS signalling function within the IMS domain (PLMN-A) also provides the LI functions when the destination party of the SMS message is a target non-local ID and, in this case, the UE-A2 may not be a target in PLMN-A.

The UE-B1 receives an SMS message over NAS. In this case the SMSF (PLMN-B) that serves the UE-B1 provides the LI functions when UE-B1 is a target in PLMN-B. The SMSF (PLMN-B) that serves the UE-B1 also provides the LI functions when the originating party of the SMS message is a target non-local ID and, in this case, the UE-B1 may not be a target in PLMN-B.

The UE-B2, an SMS over IP capable UE, receives the SMS message over NAS. In this case the SMSF (PLMN-B) that serves the UE-B2 provides the LI functions when UE-B2 is a target in PLMN-B. The SMSF (PLMN-B) that serves the

UE-B2 also provides the LI functions when the originating party of the SMS message is a target non-local ID and, in this case, the UE-B2 may not be a target in PLMN-B.

The UE-B3 receives the SMS message over IP. In this case, the IMS signalling function within the IMS domain (PLMN-B) provides the LI functions when UE-B3 is a target in PLMN-B. The IMS signalling function within the IMS domain (PLMN-B) also provides the LI functions when the origination party of the SMS message is a target non-local ID and, in this case, the UE-B3 may not be a target in PLMN-B.

4.2.2.2 UEs are roaming

Figure 4.2.2.2-1 illustrates all the possible permutations of SMS message transfer scenarios when all of the UEs happen to be roaming. In the illustration, UE-A1, UE-A2, UE-A3 belong to HPLMN-A and UE-B1, UE-B2, UE-B3, and UE-B4 belong to HPLMN-B. The UE-A1 is roaming in VPLMN-A1, UE-A2 is roaming in VPLMN-A2 (home-routed), UE-A3 is roaming in VPLMN-A3 (LBO), UE-B1 is roaming in VPLMN-B1, UE-B2 is roaming in VPLMN-B2. UE-B3 is roaming in VPLMN-B3 (home-routed), and UE-B4 is roaming in VPLMN-B4 (LBO).

The other possible network topology considerations (e.g. SMS over 4G NAS) are not in figure 4.2.2.2-1.

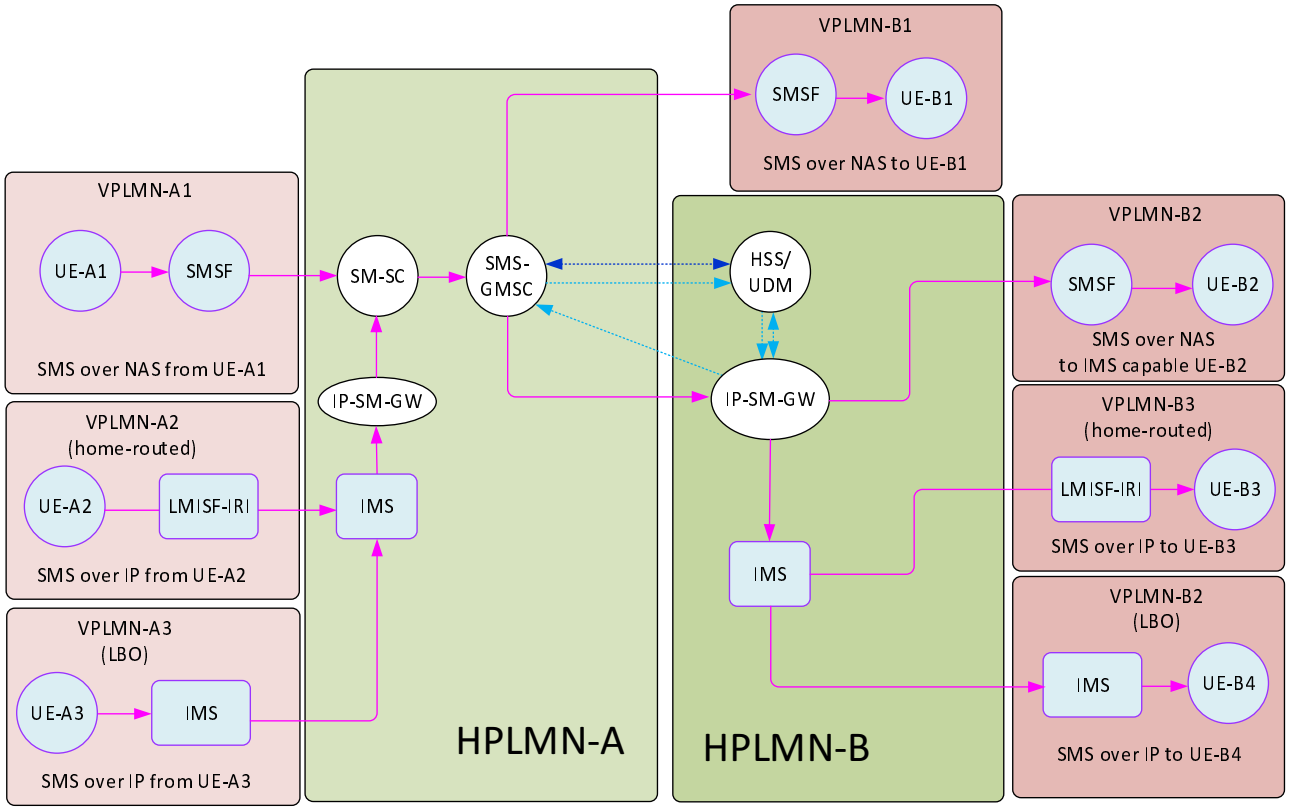


Figure 4.2.2.2-1: SMS message transfer when the UEs are roaming

The UE-A1 sends an SMS message over NAS. In this case the SMSF (VPLMN-A1) that serves the UE-A1 provides the LI functions when the UE-A1 is a target in the VPLMN-A1. The SMSF (VPLMN-A1) that serves the UE-A1 also provides the LI functions when the destination party of the SMS message is a target non-local ID and, in this case, the UE-A1 may not be a target in VPLMN-A1.

The UE-A2 sends the SMS message over IP. In this case, the LMISF-IRI (VPLMN-A2) provides the LI functions for SMS when the UE-A2 is a target in the VPLMN-A2. The IMS signalling function within the IMS domain (HPLMN-A) provides the LI functions for SMS when UE-A2 is a target in HPLMN-A. The LMISF-IRI (VPLMN-A2) and the IMS signalling function within the IMS domain (HPLMN-A) also provide the LI functions for SMS when the destination party of the SMS message is a target non-local ID and in this case, the UE-A2 may not be a target in VPLMN-A2 or HPLMN-A.

The UE-A3 sends the SMS message over IP. In this case, the IMS signalling function within the IMS domain (VPLMN-A3) provides the LI functions for SMS when the UE-A3 is a target in the VPLMN-A3. The IMS signalling function within the IMS domain (HPLMN-A) provides the LI functions for SMS when UE-A3 is a target in HPLMN-A. The

IMS signalling function within the IMS domain (VPLMN-A3) and the IMS signalling function within the IMS domain (HPLMN-A) also provide the LI functions for SMS when the destination party of the SMS message is a target non-local ID and, in this case, the UE-A3 may not be a target in VPLMN-A3 or HPLMN-A.

The UE-B1 receives an SMS message over NAS. In this case the SMSF (VPLMN-B1) that serves the UE-B1 provides the LI functions for SMS when the UE-B1 is a target in the VPLMN-B1. The SMSF (VPLMN-B1) that serves the UE-B1 also provides the LI functions when the originating party of the SMS message is a target non-local ID and, in this case, the UE-B1 may not be a target in VPLMN-B1. As shown in figure 4.2.2.2-1, the SMS messages are transferred from SMS-GMSC to the SMSF present in the VPLMN-B1.

The UE-B2 an SMS over IP capable UE receives the SMS message over NAS. In this case the SMSF (VPLMN-B2) that serves the UE-B2 provides the LI functions when UE-B2 is a target in VPLMN-B2. The SMSF (VPLMN-B2) that serves the UE-B2 also provides the LI functions when the originating party of the SMS message is a target non-local ID and, in this case, the UE-B2 may not be a target in VPLMN-B2.

The UE-B3 receives the SMS message over IP. In this case, the LMISF-IRI (VPLMN-B3) provides the LI functions for SMS when the UE-B3 is a target in VPLMN-B3. The IMS signalling function within the IMS domain (HPLMN-B) provides the LI functions for SMS when the UE-B3 is a target in HPLMN-B. The LMISF-IRI (VPLMN-B3) and the IMS signalling function within the IMS domain (HPLMN-B) also provide the LI functions when the origination party of the SMS message is a target non-local ID and, in this case, the UE-B3 may not be a target in VPLMN-B3 or HPLMN-B.

The UE-B4 receives the SMS message over IP. In this case, the IMS signalling function within the IMS domain (VPLMN-B4) provides the LI functions for SMS when the UE-B4 is a target in VPLMN-B4. The IMS signalling function within the IMS domain (HPLMN-B) provides the LI functions for SMS when the UE-B4 is a target in HPLMN-B. The IMS signalling function within the IMS domain (VPLMN-B4) and the IMS signalling function within the IMS domain (HPLMN-B) also provide the LI functions when the origination party of the SMS message is a target non-local ID and, in this case, the UE-B3 may not be a target in VPLMN-B4 or HPLMN-B.

4.2.3 SMS over NAS (UE not SMS over IP capable)

4.2.3.1 General

Figure 4.2.3.1-1 provides the generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end. Here, on both sides of the SMS message transfer, SMS over NAS is presumed. The UEs (Party_A and Party_B) are not SMS over IP capable. The UEs are not roaming.

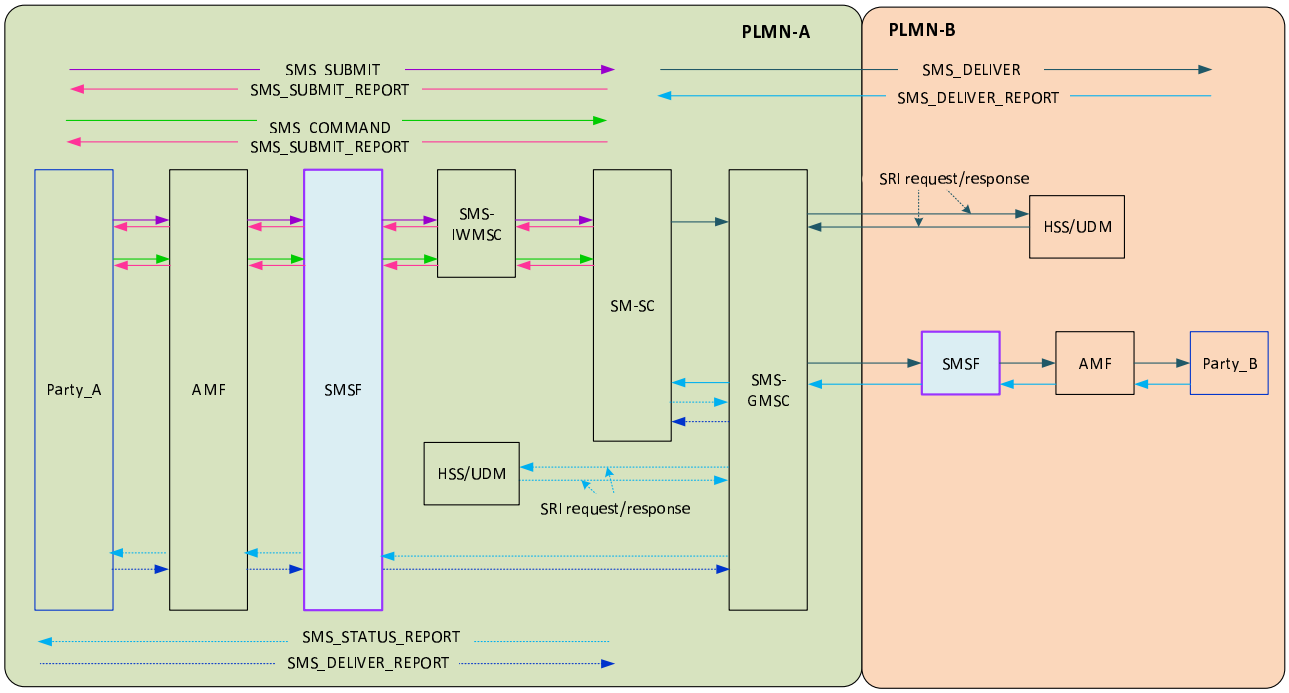


Figure 4.2.3.1-1: Generic architecture for SMS over NAS - UEs not SMS over IP capable (non-roaming)

The SMSF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in PLMN-A or when Party_B is a target non-local ID in PLMN-A. Likewise, the SMSF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in PLMN-B or when Party_A is a target non-local ID in PLMN-B.

Figure 4.2.3.1-2 provides the extreme case of a generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end where both UEs (Party_A and Party_B) happen to be roaming. Here, on both sides of the SMS message transfer, SMS over NAS is presumed. The UEs are not SMS over IP capable.

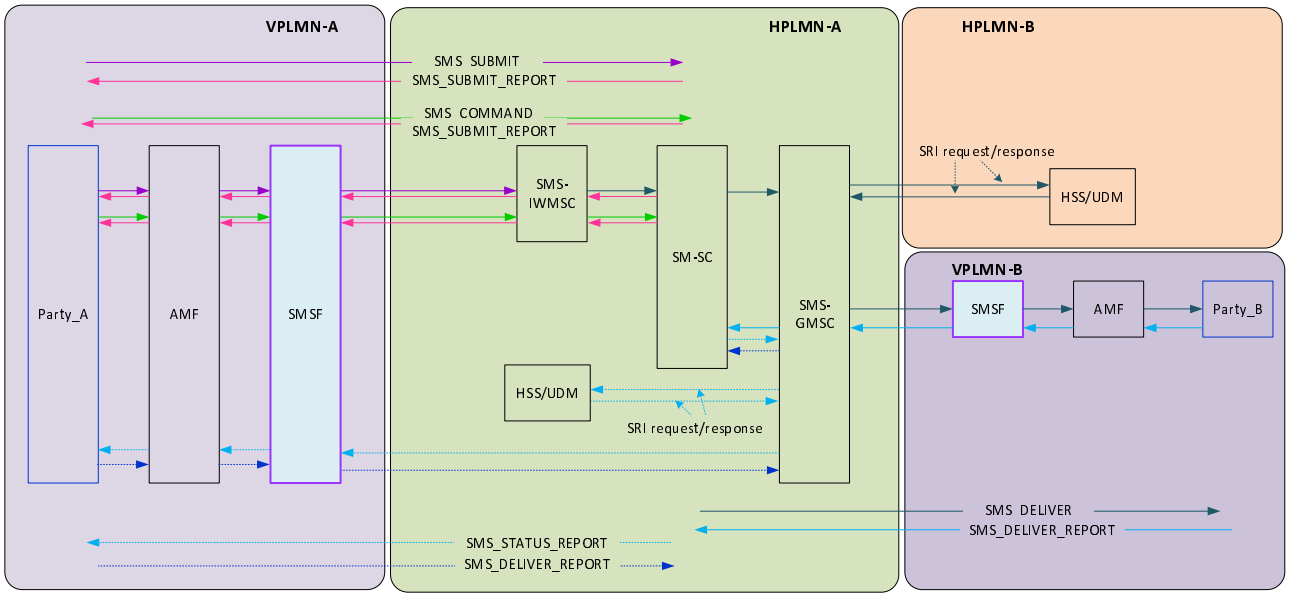


Figure 4.2.3.1-2: Generic architecture for SMS over NAS - UEs not SMS over IP capable (roaming)

The SMSF in VPLMN-A provides the IRI-POI functions when Party_A is a target in VPLMN-A or when Party_B is a target non-local ID in VPLMN-A. Likewise, the SMSF in VPLMN-B provides the IRI-POI functions when Party_B is a target in VPLMN-B or when Party_A is a target non-local ID in VPLMN-B.

NOTE: The TS 33.127 [3] and TS 33.128 [4] have not defined the LI capabilities in the HPLMN for SMS over NAS.

The scenario of Party_A is roaming, but Party_B is not roaming is possible. Likewise, the scenario of Party_B is roaming, but Party_A is not roaming is also possible. One may visualize the architectures for such scenarios easily using figures 4.2.3.1-1 and 4.2.3.1-2.

The flow-diagrams shown in the subsequent clauses are based on the architecture shown in figure 4.2.3.1-1 even though the same could be used to understand the flow-sequence when the UEs are roaming (i.e. the architecture shown in figure 4.2.3.1-2).

4.2.3.2 SMS message transfer in the originating network without a status report

The flow-diagram shown in figure 4.2.3.2-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message to the Party_B (not shown) without a status report.

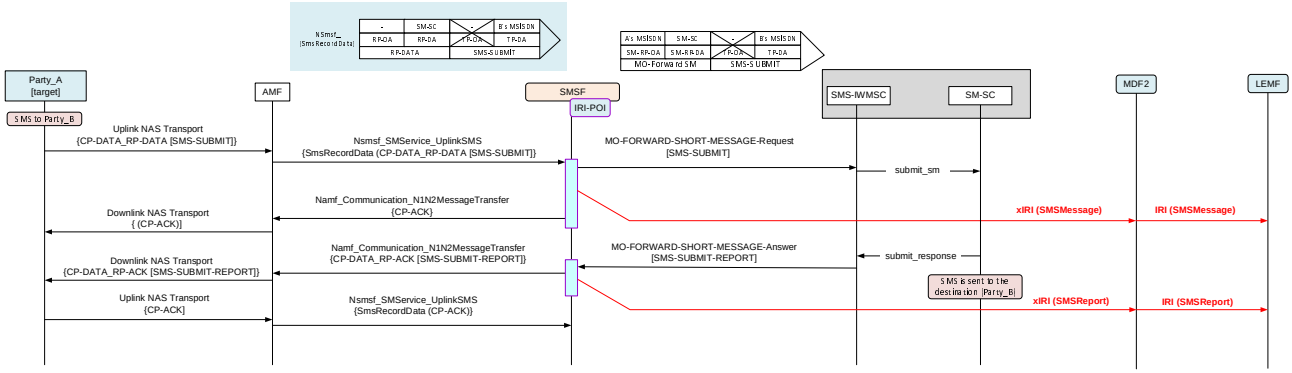


Figure 4.2.3.2-1: SMS message transfer in originating network without a status report

The illustrations shown in figure 4.2.3.2-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The GPSI, PEI or SUPI associated with the Party_A can be the target identities. The target identity determination is made from the Nsmsf and the SMSRecordData. The IRI messages SMSMessage and SMSReport have the same correlation number.

The flow-diagram shown in figure 4.2.3.2-2 illustrates the sequence of steps involved when Party_A sends an SMS message to Party_B (not shown) who happens to be a target non-local ID without a status report.

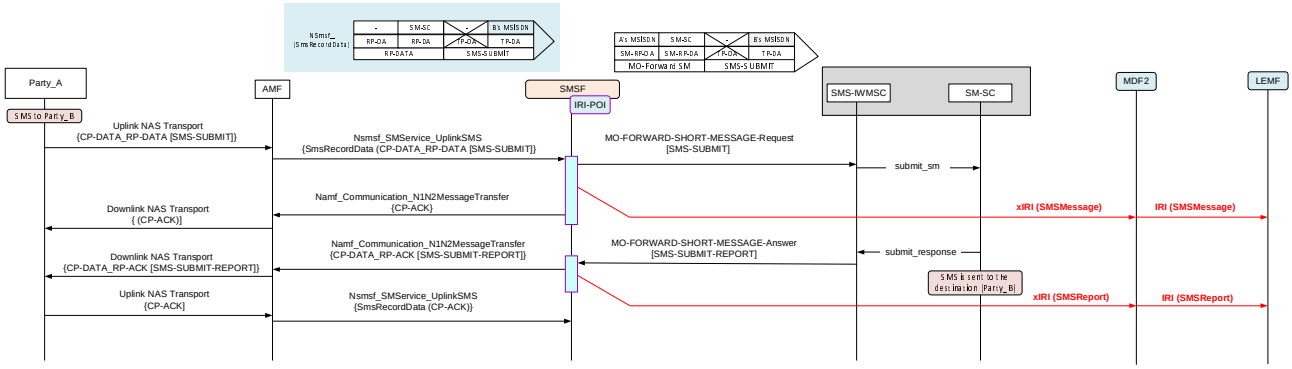


Figure 4.2.3.2-2: SMS message transfer in originating network without status report (target non-local ID)

The illustrations shown in figure 4.2.3.2-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_B (not shown) can be the identity of the target non-local ID. The target identity determination is made from the SMS-SUMBIT. The IRI messages SMSMessage and SMSReport have the same correlation number.

4.2.3.3 SMS message transfer in the originating network with a status report

The flow-diagram shown in figure 4.2.3.3-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message to the Party_B (not shown) with a status report.

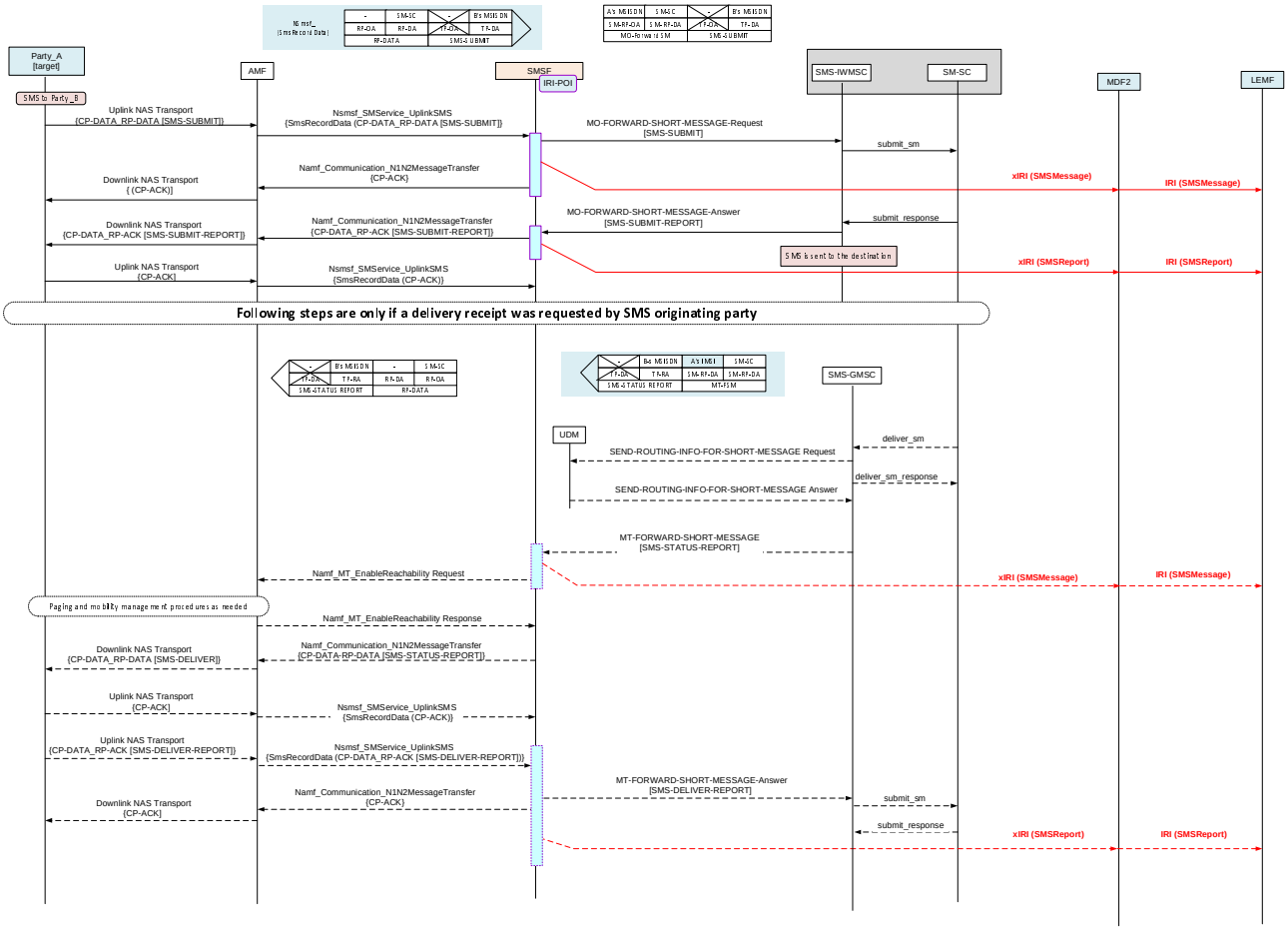


Figure 4.2.3.3-1: SMS message transfer in originating network with a status report

The illustrations shown in figure 4.2.3.3-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. For the initial xIRI SMSMessage (reporting the SMS-SUBMIT) and the associated xIRI SMSReport the GPSI, PEI or SUPI associated with the Party_A can be the target identity. The target identity determination is made from the Nsmsf and the SMSRecordData.

For the xIRI SMSMessage (reporting of the SMS-STATUS-REPORT) and the associated xIRI SMSReport, the GPSI, PEI or SUPI associated with the Party_A's TMSI or LMSI (present in the MT-Forward SM) can be the target identity.

The flow-diagram shown in figure 4.2.3.3-2 illustrates the sequence of steps involved when Party_A sends an SMS message to Party_B (not shown) who happens to be a target non-local ID with a status report.

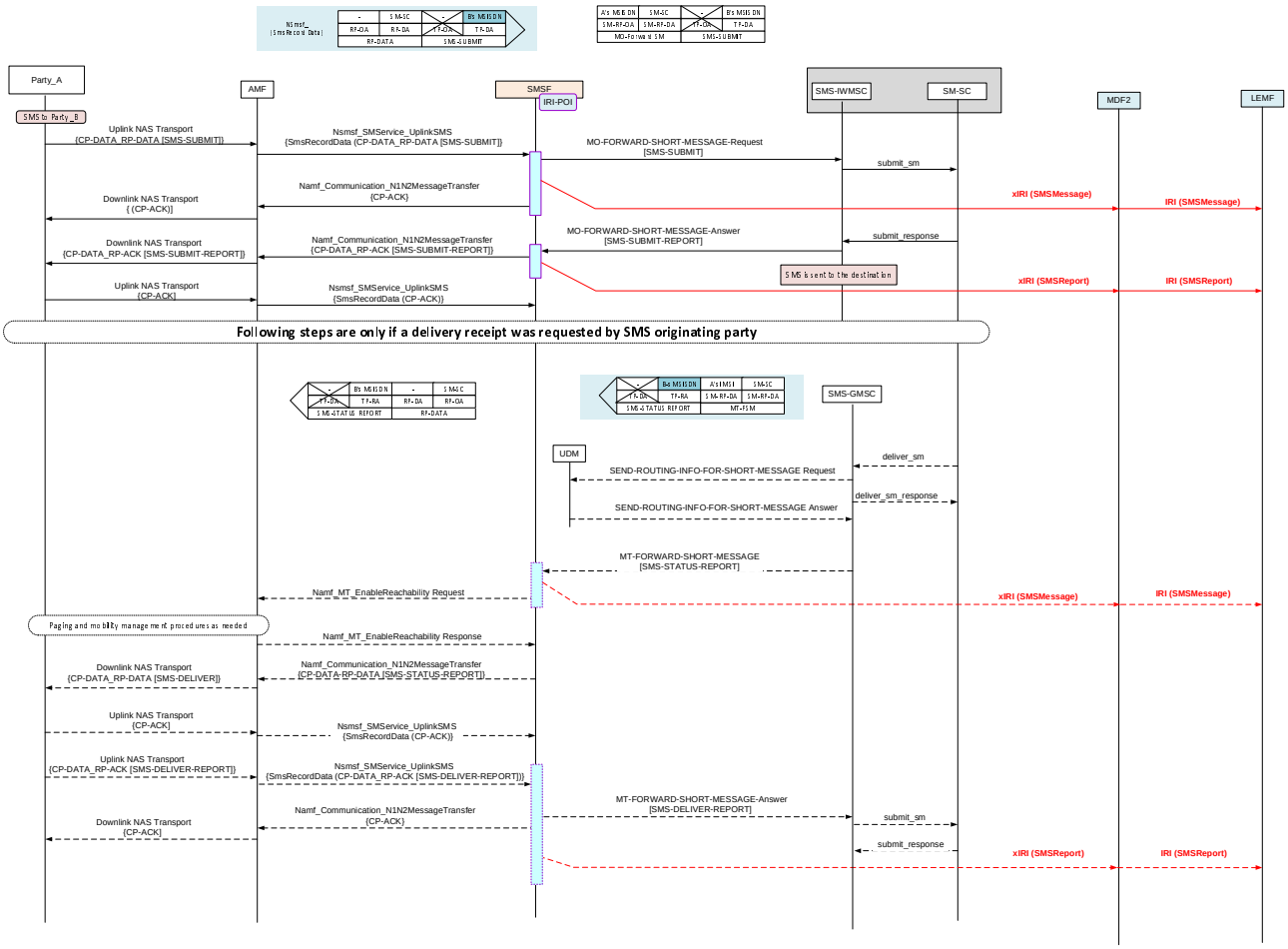


Figure 4.2.3.3-2: SMS message transfer in originating network with status report (target non-local ID)

The illustrations shown in figure 4.2.3.3-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_B (not shown) can be the identity of the target non-local ID. The target identity determination is made from the SMS-SUBMIT and the SMS-STATUS-REPORT.

In both flow diagrams shown in figures 4.2.3.3-1 and 4.2.3.3-2, the IRI messages SMSMessage for the SMS-SUBMIT and the SMSReport for the SMS-SUBMIT-REPORT have the same correlation number. Similarly, the IRI messages SMSMessage for the SMS-STATUS-REPORT and the SMSReport for the SMS-DELIVER-REPORT have the same correlation number.

4.2.3.4 SMS message transfer in the terminating network

The flow-diagram shown in figure 4.2.3.4-1 illustrates the sequence of steps involved when Party_B (target) receives an SMS message from Party_A (not shown).

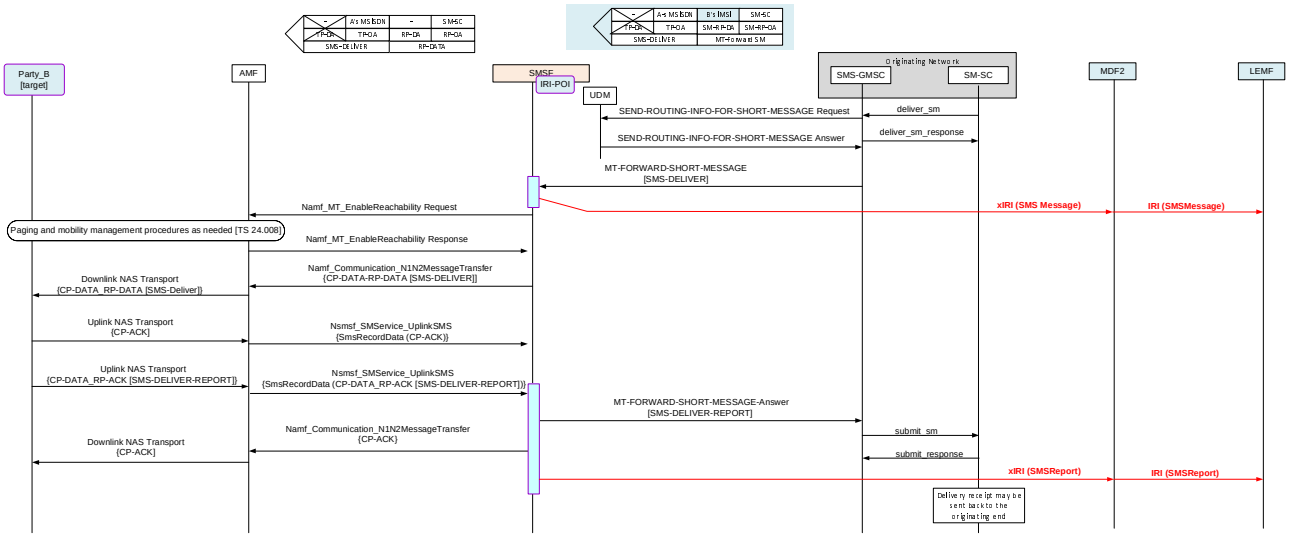


Figure 4.2.3.4-1: SMS message transfer in the terminating network

The illustrations shown in figure 4.2.3.4-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The GPSI, PEI or SUPI associated with the Party_B's TMSI or LMSI (present in the MT-Forward SM) can be the target identity. The IRI messages SMSMessage and SMSReport have the same correlation number.

The flow-diagram shown in figure 4.2.3.4-2 illustrates the sequence of steps involved when Party_B receives an SMS message from Party_A (not shown) who happens to be a target non-local ID.

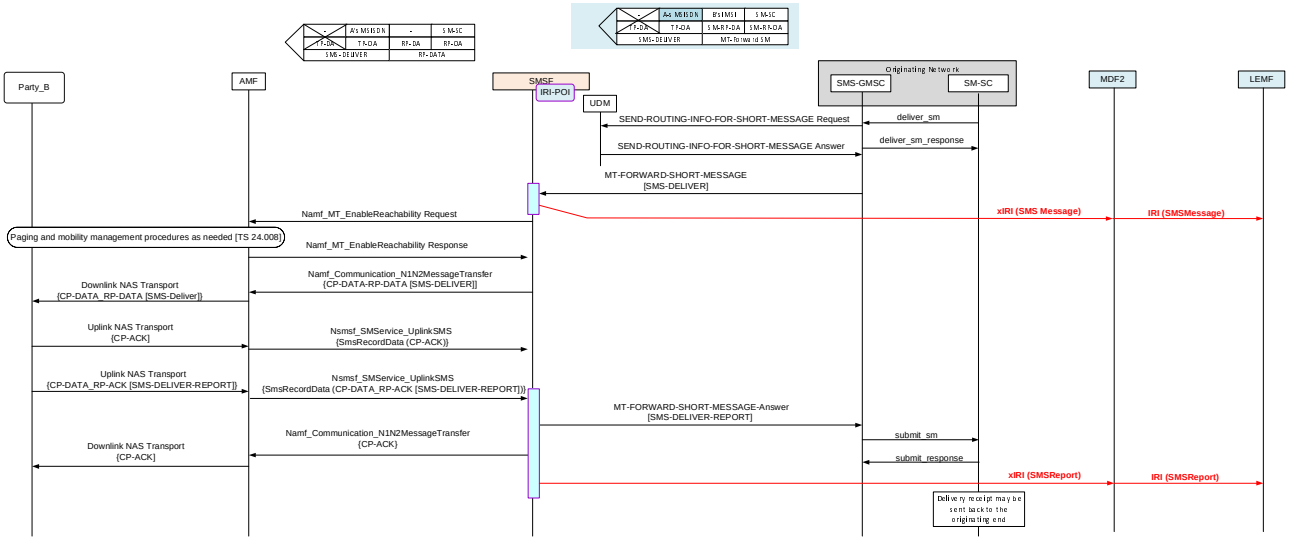


Figure 4.2.3.4-2: SMS message transfer in the terminating network (target non-local ID)

The illustrations shown in figure 4.2.3.4-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_A (not shown) can be the non-local ID. The target identity determination is made from the SMS-DELIVER. The IRI messages SMSMessage and SMSReport have the same correlation number.

4.2.3.5 Interception of SMS Command

The flow-diagram shown in figure 4.2.3.5-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS-COMMAND to the SM-SC.

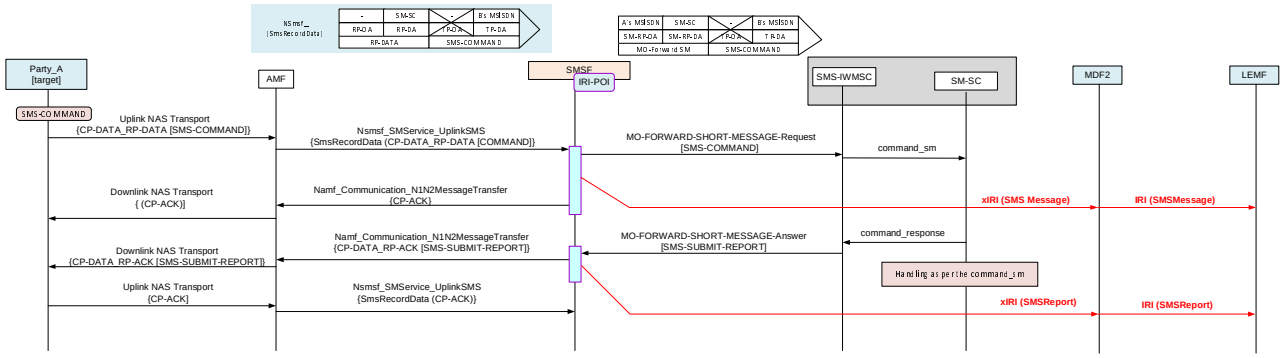


Figure 4.2.3.5-1: Interception of SMS COMMAND

The illustrations shown in figure 4.2.3.5-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The GPSI, PEI or SUPI associated with the Party_A can be the target identity. The target identity determination is made from the Nsmsf and the SMSRecordData. The IRI messages SMSMessage and SMSReport have the same correlation number.

The flow-diagram shown in figure 4.2.3.5-2 illustrates the sequence of steps involved when Party_A sends an SMS-COMMAND to the SM-SC in reference to an SMS message to Party_B (not shown) who happens to be a target non-local ID.

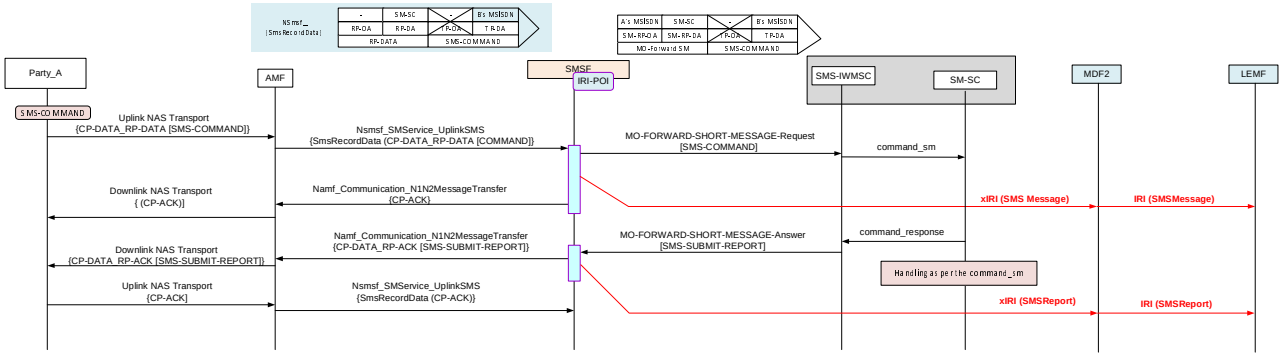


Figure 4.2.3.5-2: Interception of SMS COMMAND (target non-local ID)

The illustrations shown in figure 4.2.3.5-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_B (not shown) can be identity of the target non-local ID. The target identity determination is made from the SMS-COMMAND. The IRI messages SMSMessage and SMSReport have the same correlation number.

4.2.4 SMS over NAS (UE SMS over IP capable)

4.2.4.1 General

Figure 4.2.4.1-1 provides the generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end. Here on both side of the SMS message transfer, SMS over NAS is presumed. The UEs (Party_A and Party_B) are SMS over IP capable. The UEs are not roaming.

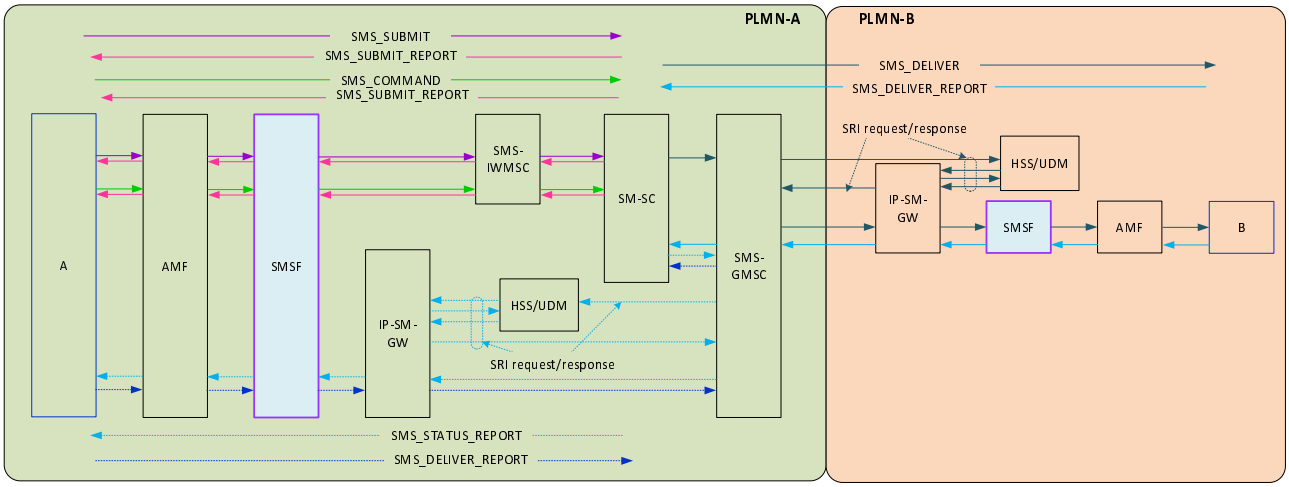


Figure 4.2.4.1-1: Generic architecture for SMS over NAS - UEs SMS over IP capable (non-roaming)

The SMSF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in PLMN-A or when Party_B is a target non-local ID in PLMN-A. Likewise, the SMSF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in PLMN-B or when Party_A is a target non-local ID in PLMN-B.

Figure 4.2.4.1-2 provides the extreme case of a generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end where both UEs (Party_A and Party_B) happen to be roaming. Here on both sides of the SMS message transfer, SMS over NAS is presumed. The UEs are SMS over IP capable.

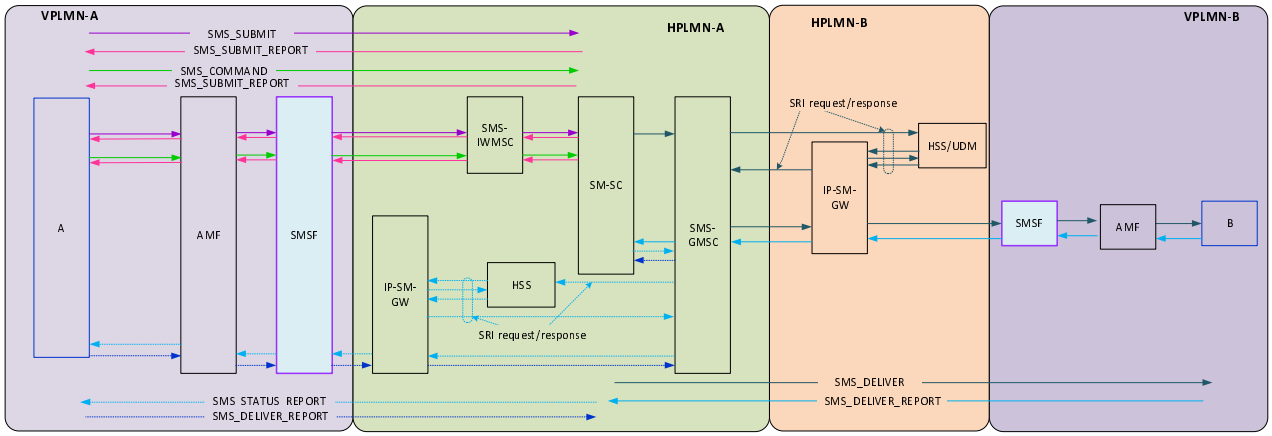


Figure 4.2.4.1-2: Generic architecture for SMS over NAS - UEs SMS over IP capable (roaming)

The SMSF in VPLMN-A provides the IRI-POI functions when Party_A is a target in VPLMN-A or when Party_B is a target non-local ID in VPLMN-A. Likewise, the SMSF in VPLMN-B provides the IRI-POI functions when Party_B is a target in VPLMN-B or when Party_A is a target non-local ID in VPLMN-B.

NOTE: The TS 33.127 [3] and TS 33.128 [4] have not defined the LI capabilities in the HPLMN for SMS over NAS.

The scenario of Party_A is roaming, but Party_B is not roaming is possible. Likewise, the scenario of Party_B is roaming, but Party_A is not roaming is also possible. One may visualize the architectures for such scenarios easily using figures 4.2.4.1-1 and 4.2.4.1-2.

The flow-diagrams shown in the subsequent clauses are based on the architecture shown in figure 4.2.4-1 even though the same could be used to understand the flow-sequence when the UEs are roaming (i.e. architecture shown in figure 4.2.4.1-2).

4.2.4.2 SMS message transfer in the originating network without a status report

The flows are the same as the flow-diagrams illustrated in figure 4.2.3.2-1 and 4.2.3.2-2 (see clause 4.2.3.2).

4.2.4.3 SMS message transfer in the terminating network

The flow-diagram shown in figure 4.2.4.3-1 illustrates the sequence of steps involved when Party_B (target) receives an SMS message from Party_A (not shown).

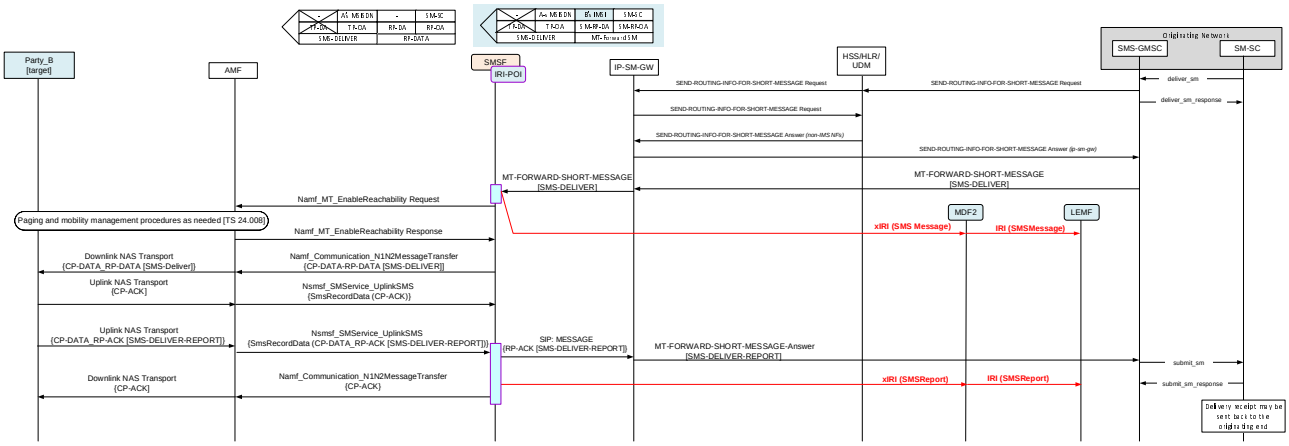


Figure 4.2.4.3-1: SMS message transfer in the terminating network

The illustrations shown in figure 4.2.4.3-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The GPSI, PEI or SUPI associated with the Party_B's TMSI or LMSI (present in the MT-Forward SM) can be the target identity. The IRI messages SMSMessage and SMSReport have the same correlation number.

The flow-diagram shown in figure 4.2.4.3-2 illustrates the sequence of steps involved when Party_B receives an SMS message from Party_A (not shown) who happens to be a target non-local ID.

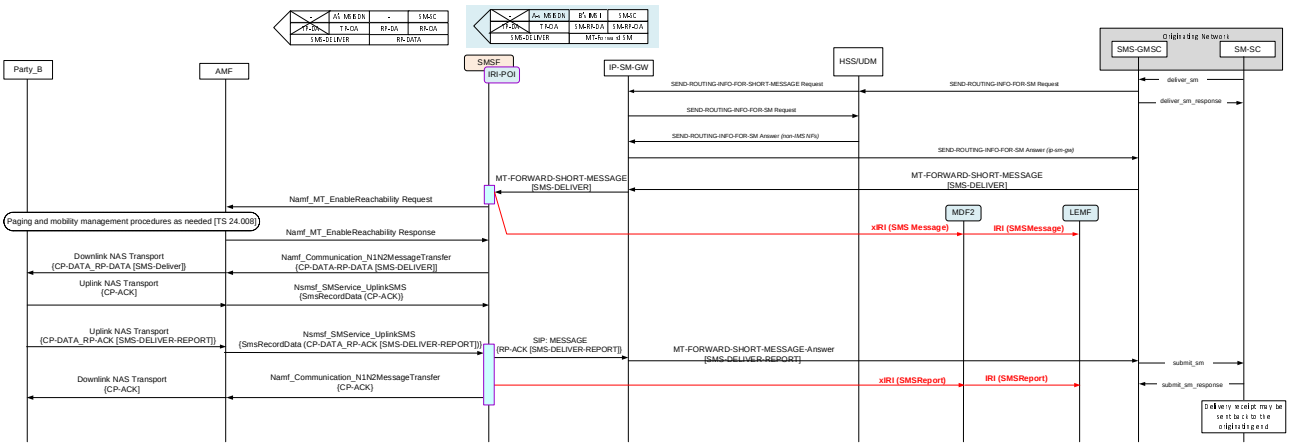


Figure 4.2.4.3-2: SMS message transfer in the terminating network (target non-local ID)

The illustrations shown in figure 4.2.4.3-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_A (not shown) can be the non-local ID. The target identity determination is made from the SMS-DELIVER. The IRI messages SMSMessage and SMSReport have the same correlation number.

When the HSS receives a query from the SMS-GMSC, since the terminating UE is SMS over IP capable, the HSS forwards the request to the IP-SM-GW. The IP-SM-GW sends the query to the HSS to determine the non-IMS NFs that can handle the SMS for that UE. The HSS returns those non-IMS NF addresses, in this flow, it is that of the SMSF. Then, the IP-SM-GW returns its own address to the SMS-GMSC. The SMS-GMSC, that way sends the SMS to the IP-

SM-GW and the IP-SM-GW routes the SMS to the UE depending on to which NF the UE is currently registered to. In this flow, it is the SMSF.

4.2.4.4 SMS message transfer in the originating network with a status report

The flow-diagram shown in figure 4.2.4.4-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message to the Party_B (not shown) with a status report.

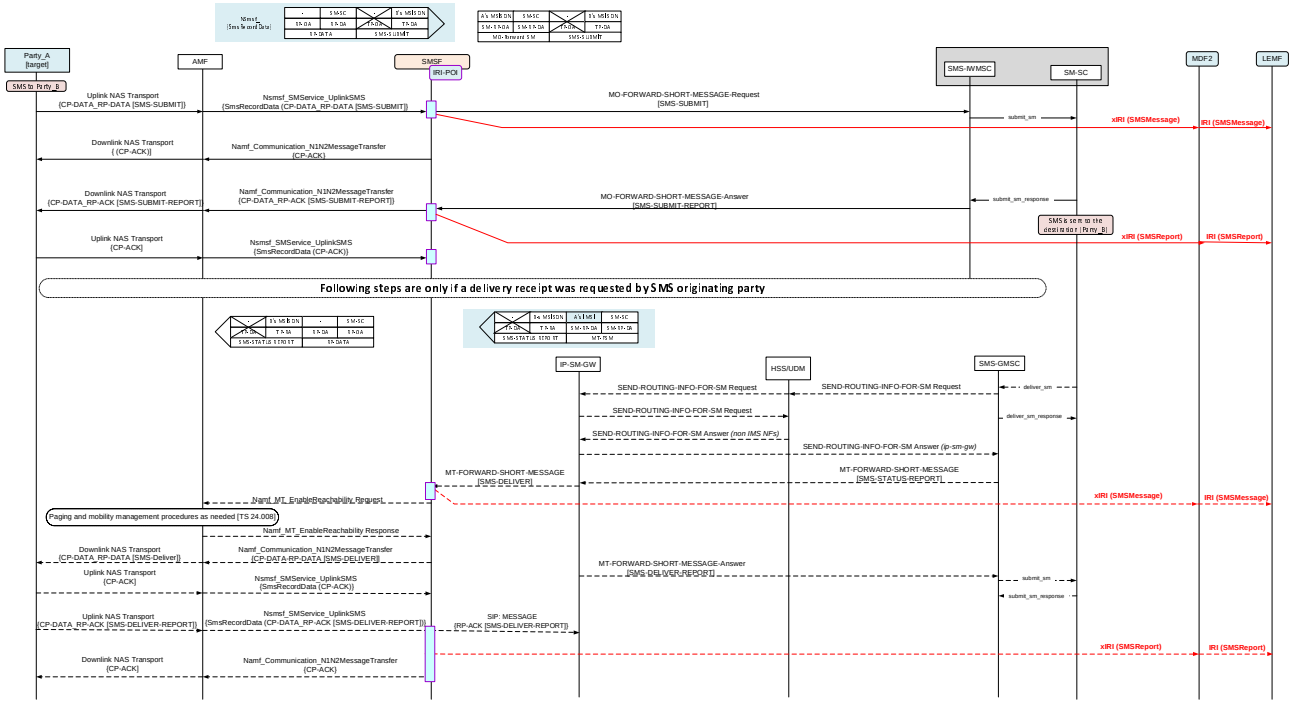


Figure 4.2.4.4-1: SMS message transfer in originating network with a status report

The illustrations shown in figure 4.2.4.4-1 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. For the initial xIRI SMSMessage (reporting the SMS-SUBMIT) and the associated xIRI SMSReport the GPSI, PEI or SUPI associated with the Party_A can be the target identity. The target identity determination is made from the Nsmsf and the SMSRecordData.

For the xIRI SMSMessage (reporting of SMS-STATUS-REPORT) and the associated xIRI SMSReport, the GPSI, PEI or SUPI associated with the A's TMSI or LMSI (present in the MT-Forward SM) can be the target identity.

The flow-diagram shown in figure 4.2.4.4-2 illustrates the sequence of steps involved when Party_A sends an SMS message to Party_B (not shown) who happens to be a target non-local ID with a status report.

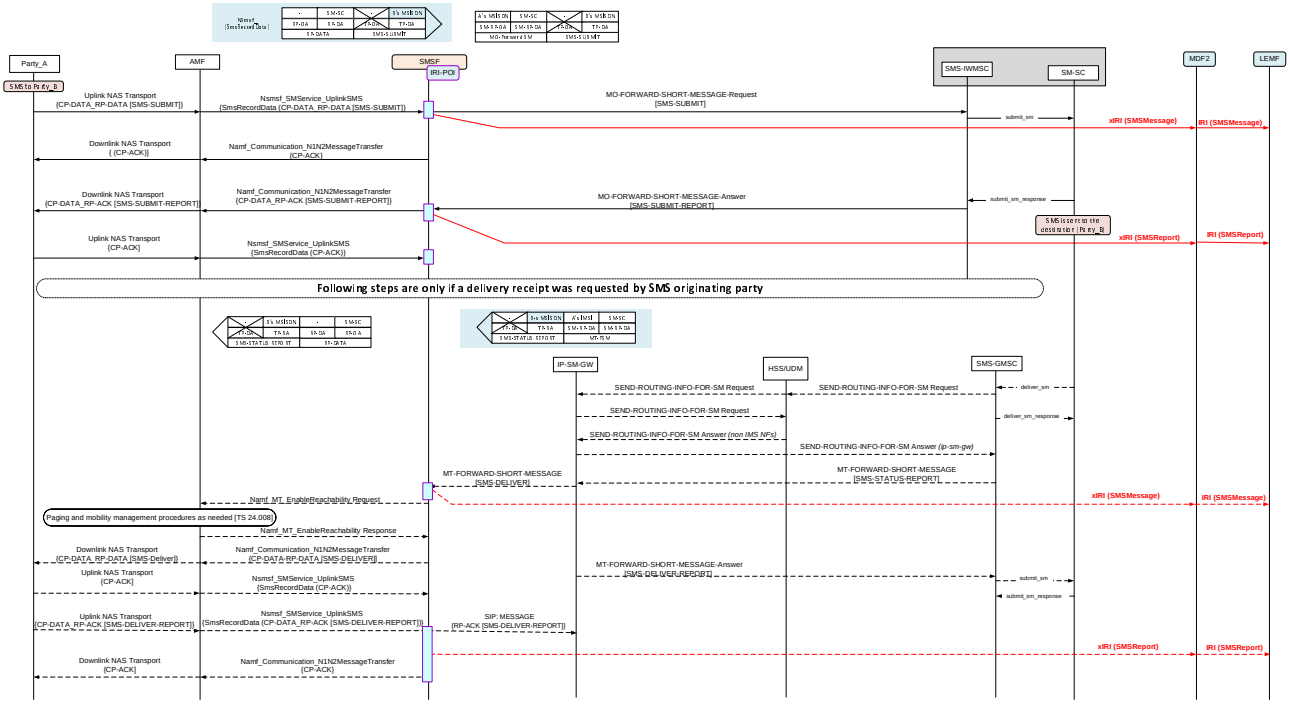


Figure 4.2.4.4-2: SMS message transfer in originating network with status report (target non-local ID)

The illustrations shown in figure 4.2.4.4-2 are based on the event triggers defined in TS 33.128 [4] for the generation of the xIRI SMSMessage and the xIRI SMSReport. The MSISDN associated with the Party_B (not shown) can be identity of the target non-local ID. The target identity determination is made from the SMS-SUBMIT and the SMS-STATUS-REPORT.

In both flow diagrams shown in figure 4.2.4.4-1 and 4.2.4.4-2, the IRI messages SMSMessage for the SMS-SUBMIT and the SMSReport for the SMS-SUBMIT-REPORT have the same correlation number. Similarly, the IRI messages SMSMessage for the SMS-STATUS-REPORT and the SMSReport for the SMS-DELIVER-REPORT have the same correlation number.

4.2.4.5 Interception of SMS Command

The flows are same as the flow-diagrams illustrated in figure 4.2.3.5-1 and 4.2.3.5-2 (see clause 4.2.3.5).

4.2.5 SMS over IP

4.2.5.1 General

4.2.5.1.1 Possible variations

The following three methods are presented to illustrate the LI aspects for SMS over IP:

- SMS over IP with MSISDN as the destination.
- SMS over IP with SIP URI as the destination.
- Instant messaging mapped to SMS.

In the first two cases, the UE sends the SIP MESSAGE that includes "vnd.3gpp.sms" as the MIME content type in the payload with RP-DATA or RP-ACK as the content (TS 24.341 [5]. In the second scenario, the SIP MESSAGE additionally also includes "vnd.3gpp.sms+xml" as MIME content type in the payload (TS 24.341 [5].

In the third scenario, the UE sends a SIP MESSAGE that does not include the "vnd.3gpp.sms" as the MIME content type, however, since the instant messaging is mapped to a SMS, the terminating end of the message transfer includes the "vnd.3gpp.sms" as the content type in the payload before delivering the same to the destination.

4.2.5.1.2 SMS over IP with MSISDN as the destination

4.2.5.1.2.1 Topology views

Figure 4.2.5.1.2.1-1 provides the generic architecture showing major NFs involved in the transfer of SMS messages from the originating end to the terminating end. Here on both side of the SMS message transfer, SMS over IP is presumed. The UEs are not roaming.

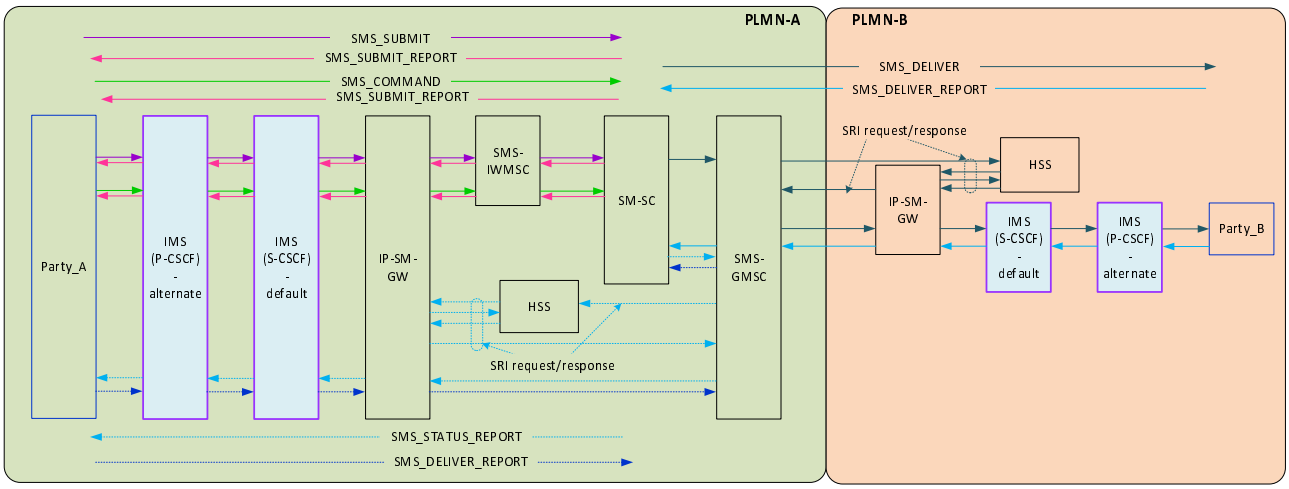


Figure 4.2.5.1.2.1-1: Generic architecture for SMS over IP (non-roaming) with MSISDN as the destination

In the default deployment option (see TS 33.127 [3] and TS 33.128 [4]), the S-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in PLMN-A or when Party_B is a target non-local ID in PLMN-A. Likewise, the S-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in PLMN-B or when Party_A is a target non-local ID in PLMN-B.

In the alternate deployment option (see TS 33.127 [3] and TS 33.128 [4]), the P-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in PLMN-A or when Party_B is a target non-local ID in PLMN-A. Likewise, the P-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in PLMN-B or when Party_A is a target non-local ID in PLMN-B.

Figure 4.2.5.1.2.1-2 provides the extreme case of a generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end where both UEs (Party_A and Party_B) happen to be roaming using the home-routed architecture. Here on both side of the SMS message transfer, SMS over IP is presumed.

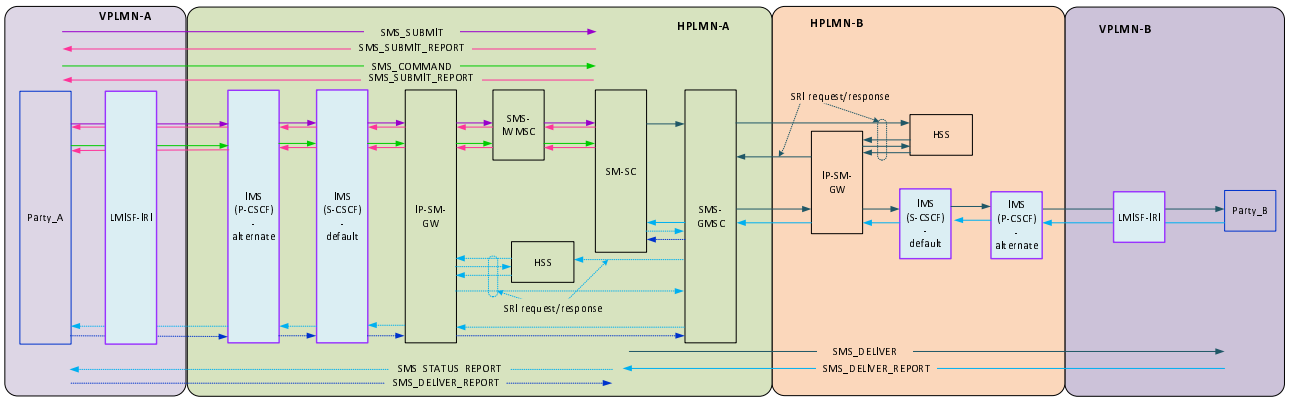


Figure 4.2.5.1.2.1-2: Generic architecture for SMS over IP (roaming with home-routed) with MSISDN as the destination

The LMISF-IRI in the VPLMN-A provides the IRI-POI functions when Party_A is a target in the VPLMN-A or when Party_B is a target non-local ID in the VPLMN-A. Likewise, the LMISF-IRI in the VPLMN-B provides the IRI-POI functions when Party_B is a target in the VPLMN-B or when Party_A is a target non-local ID in the VPLMN-B.

In the default deployment option (see TS 33.127 [3] and TS 33.128 [4]), the S-CSCF in the HPLMN-A provides the IRI-POI functions when Party_A is a target in the HPLMN-A or when Party_B is a target non-local ID in the HPLMN-A. Likewise, the S-CSCF in the HPLMN-B provides the IRI-POI functions when Party_B is a target in the HPLMN-B or when Party_A is a target non-local ID in the HPLMN-B.

In the alternate deployment option (see TS 33.127 [3] and TS 33.128 [4]), the P-CSCF in the HPLMN-A provides the IRI-POI functions when Party_A is a target in the HPLMN-A or when Party_B is a target non-local ID in the HPLMN-A. Likewise, the P-CSCF in the HPLMN-B provides the IRI-POI functions when Party_B is a target in the HPLMN-B or when Party_A is a target non-local ID in the HPLMN-B.

Figure 4.2.5.1.2.1-3 provides the extreme case of generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end where both UEs (Party_A and Party_B) happen to be roaming with LBO. Here on both sides of the SMS message transfer, SMS over IP is presumed.

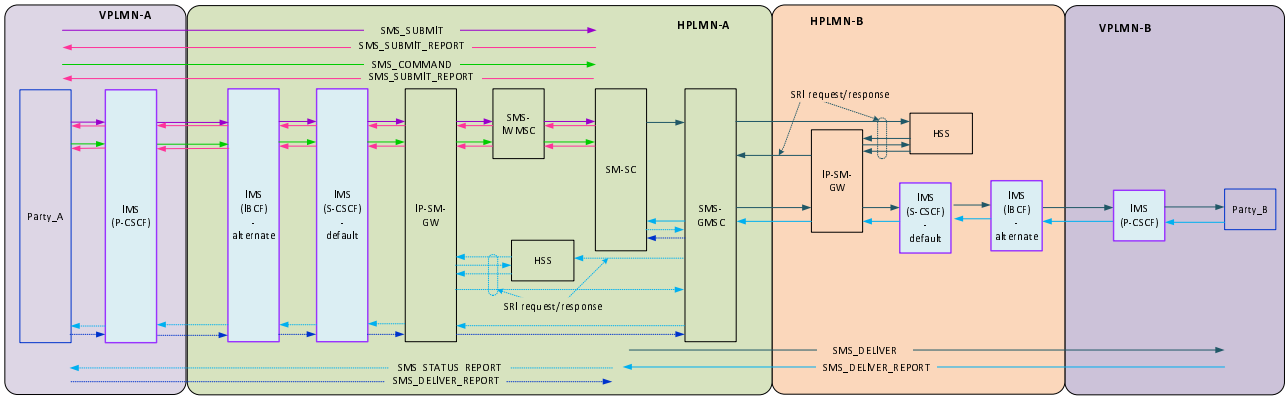


Figure 4.2.5.1.2.1-3: Generic architecture for SMS over IP (roaming with LBO) with MSISDN as the destination

The P-CSCF in VPLMN-A provides the IRI-POI functions when Party_A is a target in VPLMN-A or when Party_B is a target non-local ID in the VPLMN-A. Likewise, the P-CSCF in the VPLMN-B provides the IRI-POI functions when Party_B is a target in the VPLMN-B or when Party_A is a target non-local ID in the VPLMN-B.

In the default deployment option (see TS 33.127 [3] and TS 33.128 [4]), the S-CSCF in the HPLMN-A provides the IRI-POI functions when Party_A is a target in the HPLMN-A or when Party_B is a target non-local ID in the HPLMN-A. Likewise, the S-CSCF in the HPLMN-B provides the IRI-POI functions when Party_B is a target in the HPLMN-B or when Party_A is a target non-local ID in the HPLMN-B.

In the alternate deployment option (see TS 33.127 [3] and TS 33.128 [4]), the IBCF in the HPLMN-A provides the IRI-POI functions when Party_A is a target in the HPLMN-A or when Party_B is a target non-local ID in the HPLMN-A. Likewise, the IBCF in the HPLMN-B provides the IRI-POI functions when Party_B is a target in the HPLMN-B or when Party_A is a target non-local ID in the HPLMN-B.

The scenario of Party_A is roaming, but Party_B is not roaming is possible. Likewise, the scenario of Party_B is roaming, but Party_A is not roaming is also possible. One may visualize the architectures for such scenarios easily using figures 4.2.5.1.2.1-1, 4.2.5.1.2.1-2, 4.2.5.1.2.1-3.

4.2.5.1.2.2 Flow diagrams

4.2.5.1.2.2.1 General

As defined in TS 33.128[4], all the applicable SIP messages are reported from the IRI-POI using the xIRI IMSMessage. Therefore, the SIP MESSAGE that includes the SMS (within the payload of content type "vnd.3gpp.sms") are also reported from the IRI-POI as the xIRI IMSMessage. This is different from the SMS over NAS where the SMS messages are reported from the IRI-POI using the xIRI SMSMessage.

The details of how the IRI-POI performs a target match is described in TS 33.128 [4] clause 4.12.2.8.2.3.

The non-LI aspect of the flow diagrams shown in the subsequent clauses are based on the flow diagrams defined in 3GPP TS 24.341 [5].

4.2.5.1.2.2.2 Originating network without status report

The flow-diagram shown in figure 4.2.5.1.2.2.2-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message over IP to the Party_B (not shown) without a status report.

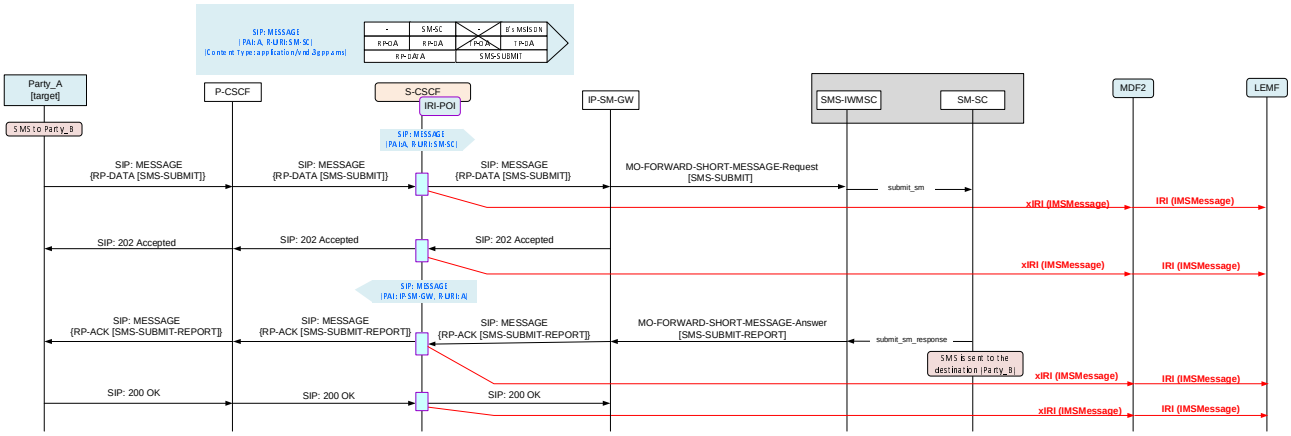


Figure 4.2.5.1.2.2.2-1: SMS over IP message transfer in originating network without a status report

The IRI messages in the above flow-diagram have the same correlation number.

The P-Asserted Identity and From fields of the SIP MESSAGE that includes the SMS-SUBMIT have the A party identity (that happens to be the target in this flow). The Request URI and the To field of the SIP MESSAGE that includes the SMS-SUBMIT -REPORT have the A party identity (which happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.2.2.2-2 illustrates the sequence of steps involved when Party_A sends an SMS message over IP to Party_B (not shown) who happens to be a target non-local ID without a status report.

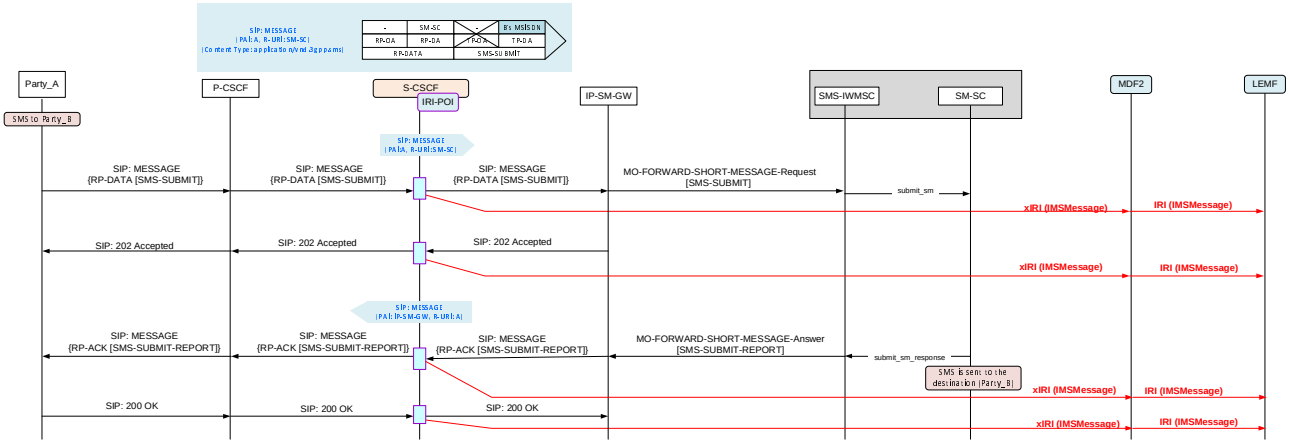


Figure 4.2.5.1.2.2.2-2: SMS over IP message transfer in originating network without a status report (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The TP-DA field of the SMS-SUBMIT includes the destination number (that happens to be the target non-local ID in this flow). The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-SUBMIT-REPORT later sent back the A-party is also intercepted.

4.2.5.1.2.2.3 Originating network with a status report

The flow-diagram shown in figure 4.2.5.1.2.2.3-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message over IP to the Party_B (not shown) with a status report.

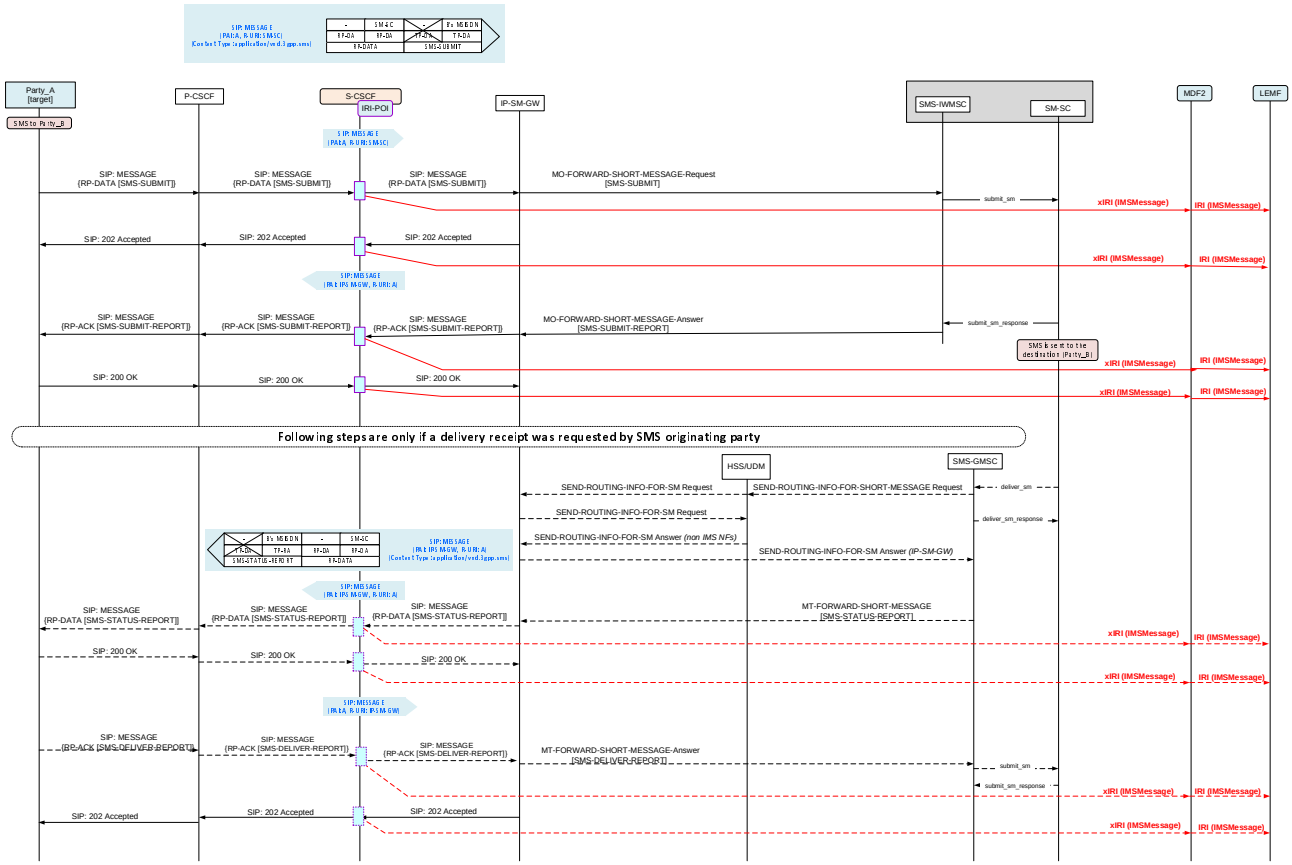


Figure 4.2.5.1.2.2.3-1: SMS over IP message transfer in originating network with a status report

The IRI messages in the above flow-diagram have the same correlation number.

The first part of this flow is same as the flow shown in figure 4.2.5.1.2.2.2-1. The Request URI and the To fields of the SIP MESSAGE that includes the SMS-STATUS-REPORT have the A-party identity (that happens to be the target in this flow). The P-Asserted Identity and the From fields of SIP MESSAGE that includes the SMS-DELIVER -REPORT have the A party identity (which happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.2.2.3-2 illustrates the sequence of steps involved when Party_A sends an SMS message over IP to Party_B (not shown) who happens to be a target non-local ID with a status report.

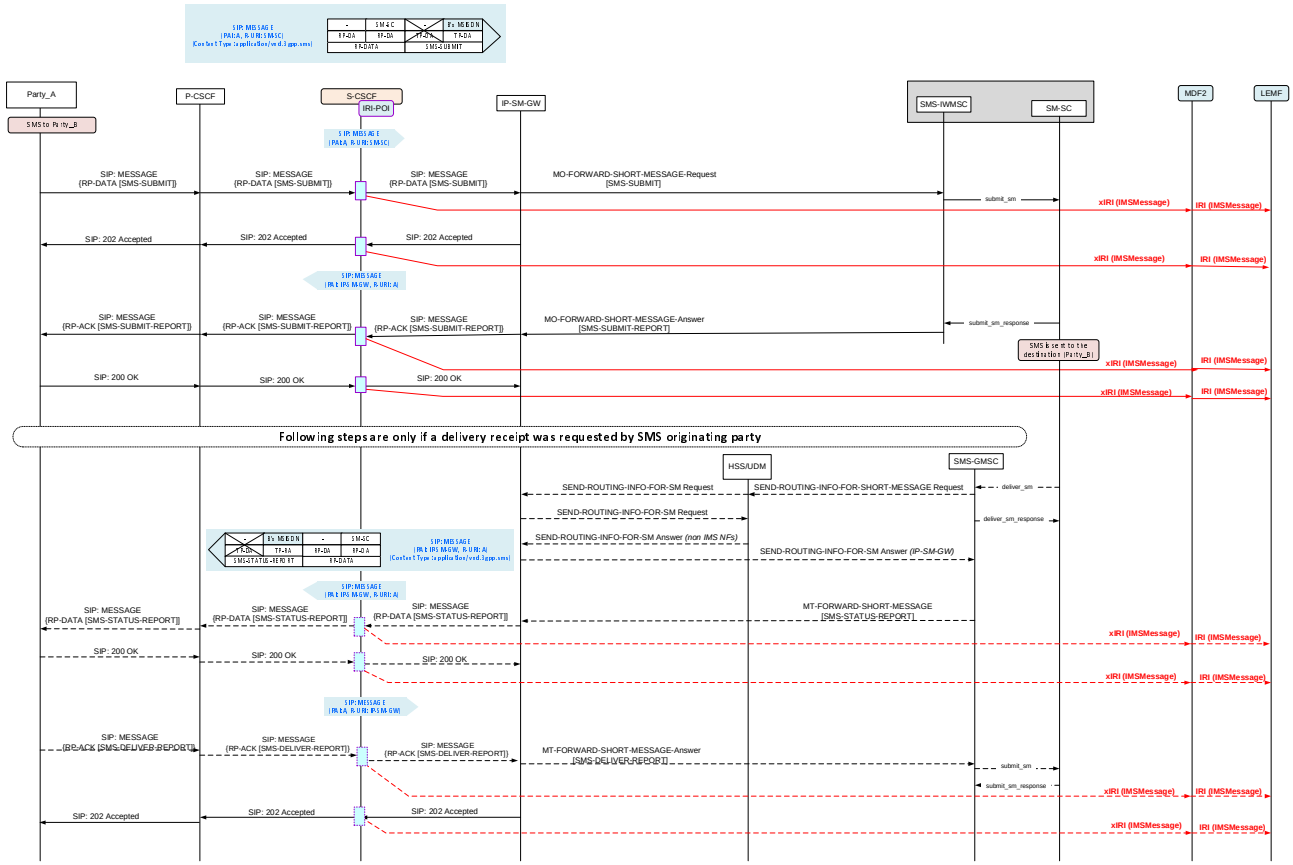


Figure 4.2.5.1.2.2.3-2: SMS over IP message transfer in originating network with a status report (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The first part of this flow is same as flow shown in figure 4.2.5.1.2.2.3-1. The TP-RA field of SMS-STATUS-REPORT include the destination number (that happens to be the target non-local ID in this flow) of the associated SMS message sent by the A-party. The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-DELIVER-REPORT later sent back the SMS-GMSC is also intercepted.

4.2.5.1.2.2.4 Terminating network – MSISDN as the destination

The flow-diagram shown in figure 4.2.5.1.2.2.4-1 illustrates the sequence of steps involved when Party_B (target) receives an SMS message from Party_A (not shown).

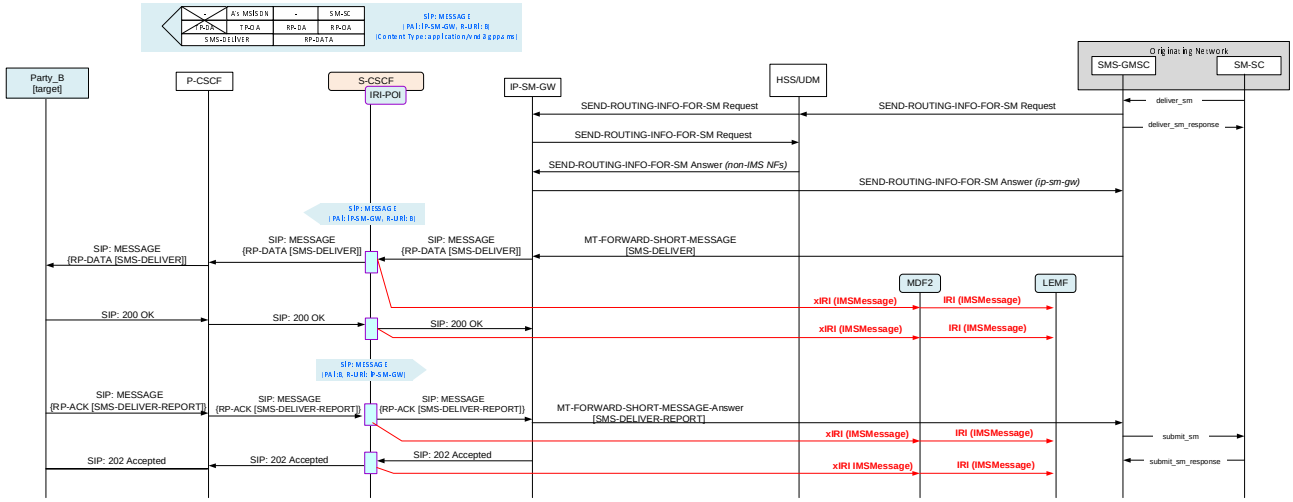


Figure 4.2.5.1.2.2.4-1: SMS over IP message transfer in terminating network

The IRI messages in the above flow-diagram have the same correlation number.

The Request URI and the To fields of the SIP MESSAGE that includes the SMS-DELIVER have the B party identity (that happens to be the target in this flow). The P-Asserted Identity and the From fields of the SIP MESSAGE that includes the SMS-DELIVER -REPORT have the B party identity (that happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.2.2.4-2 illustrates the sequence of steps involved when Party_B receives an SMS message from Party_A (not shown) who happens to be a target non-local ID.

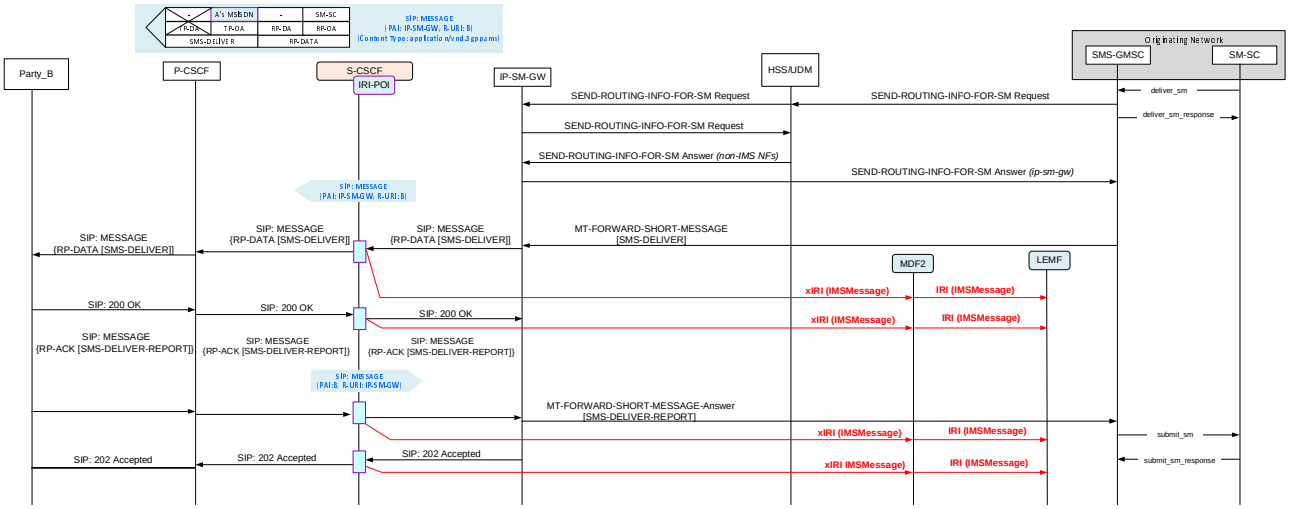


Figure 4.2.5.1.2.2.4-2: SMS over IP message transfer in terminating network (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The TP-OA field of SMS-DELIVER includes the origination number (that happens to the target non-local ID in this flow). The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-DELIVER-REPORT later sent back the SMS-GMSC is also intercepted.

4.2.5.1.2.2.5 Interception of SMS Command

The flow-diagram shown in figure 4.2.5.1.2.2.5-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS-COMMAND to the SM-SC.

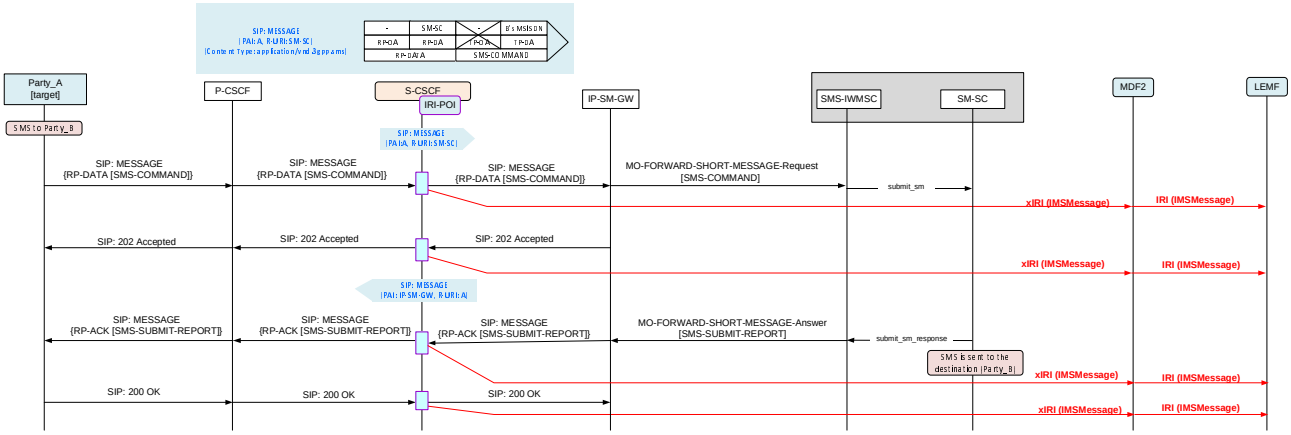


Figure 4.2.5.1.2.2.5-1: Interception of SMS COMMAND

The IRI messages in the above flow-diagram have the same correlation number.

The P-Asserted Identity and From fields of the SIP MESSAGE that includes the SMS-COMMAND have the A party identity (that happens to be the target in this flow). The Request URI and the To fields of the SIP MESSAGE that includes the SMS-SUBMIT -REPORT have the A party identity (which happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.2.2.5-2 illustrates the sequence of steps involved when Party_A sends an SMS-COMMAND to the SM-SC in reference to an SMS message to Party_B (not shown) who happens to be a target non-local ID.

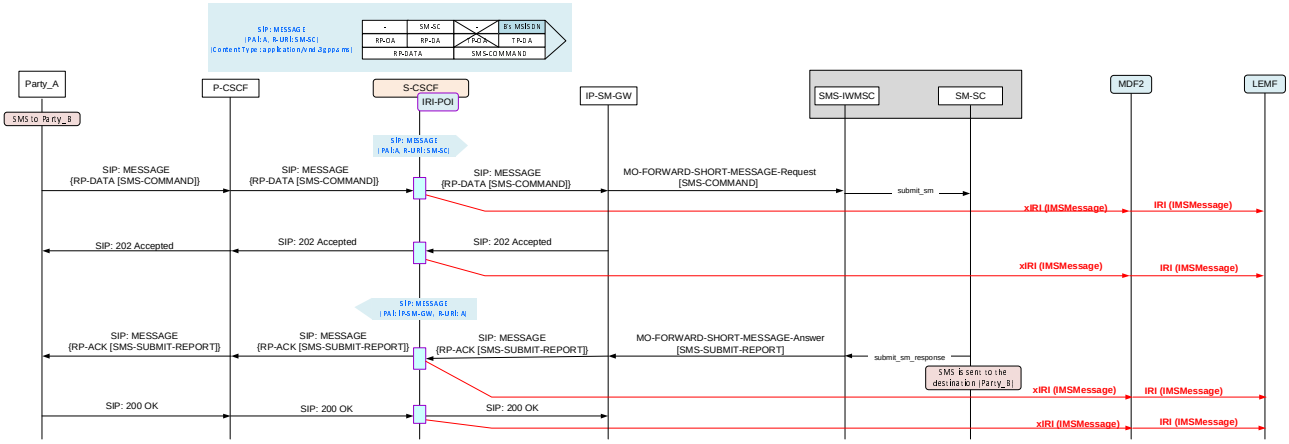


Figure 4.2.5.1.2.2.5-2: Interception of SMS COMMAND (destination is target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The TP-DA field of the SMS-COMMAND includes the destination number (that happens to be the target non-local ID in this flow). The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-SUBMIT-REPORT later sent back the A-party is also intercepted.

4.2.5.1.3 SMS over IP with SIP URI as the destination

4.2.5.1.3.1 Topology views

Figure 4.2.5.1.3.1-1 provides the generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end when the SIP IRI is used to address the destination. Here on both sides of the SMS message transfer, SMS over IP is presumed. The UEs are not roaming.

Unlike the normal SMS over IP (where the destination is addressed via MSISDN), when the SIP URI is the destination, both originating end and terminating end will have the SMS over IP. With the SMS over IP (MSISDN as the destination), mixed permutations (e.g. originating side with SMS over IP, terminating side with SMS over NAS) are possible.

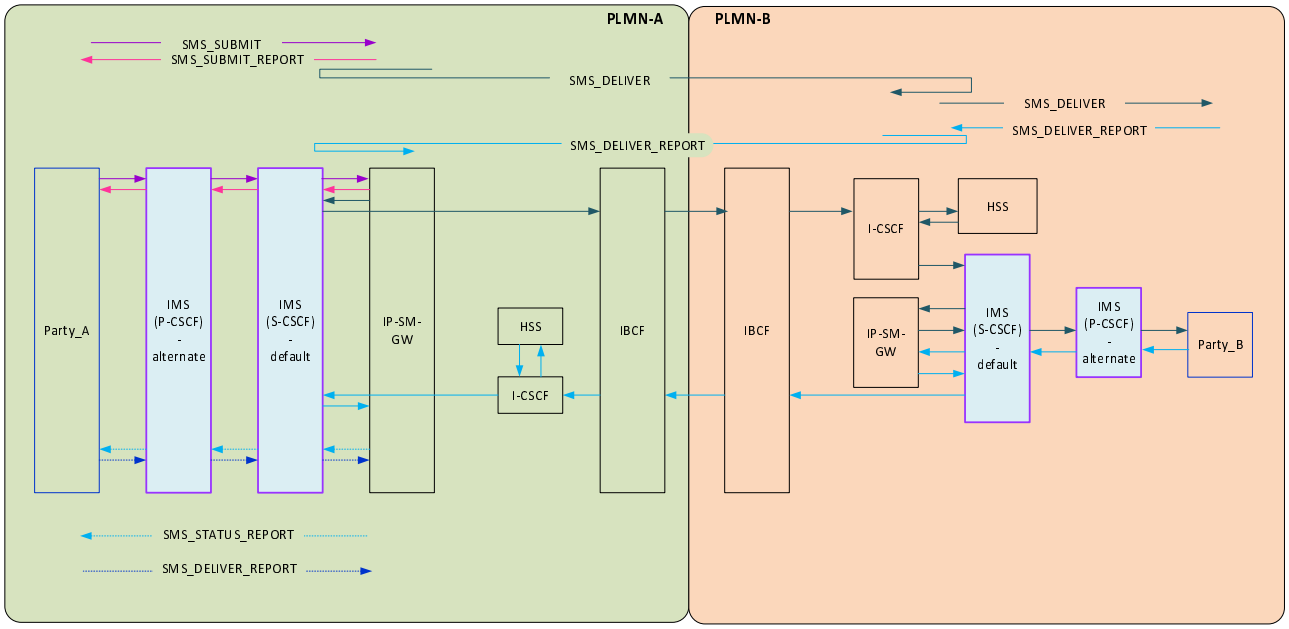


Figure 4.2.5.1.3.1-1: Generic architecture for SMS over IP (not roaming) with SIP URI as the destination

As shown in figure 4.2.5.1.3.1-1, the SM-SC is not involved in the message transfer path because the destination is addressed using a SIP URI.

For the LI, in the default deployment option (see TS 33.127 [3] and TS 33.128 [4]), the S-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in the PLMN-A or when Party_B is a target non-local ID in the PLMN-A. Likewise, the S-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in the PLMN-B or when Party_A is a target non-local ID in the PLMN-B.

In the alternate deployment option (see TS 33.127 [3] and TS 33.128 [4]), the P-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in the PLMN-A or when Party_B is a target non-local ID in the PLMN-A. Likewise, the P-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in the PLMN-B or when Party_A is a target non-local ID in the PLMN-B.

As can be observed in figures 4.2.5.1.2.1-1 and 4.2.5.1.3.1-1, the NFs involved in the message transfer from the originating UE to the IP-SM-GW and from the terminating IP-SM-GW to the termination UE are the same for the two cases. Therefore, no separate diagrams are used to illustrate the roaming scenarios of SMS over IP with SIP URI as the destination. The illustrations shown in figures 4.2.5.1.2.1-2, and 4.2.5.1.2.1-3 may be used to understand the topology views when the UEs are roaming (Home-Routed or LBO).

4.2.5.1.3.2 Flow diagrams

4.2.5.1.3.2.1 General

As defined in TS 33.128[4], all the applicable SIP messages are reported from the IRI-POI using the xIRI IMSMessage. Therefore, the SIP MESSAGE that includes the SMS (within the payload of content type "vnd.3gpp.sms") are reported from the IRI-POI as the xIRI IMSMessage. This is different from the SMS over NAS where the SMS messages are reported from the IRI-POI using the xIRI SMSMessage.

The SIP MEESAGE that includes the SMS when destination is addressed using a SIP URI also includes another MIME content type "vnd.3gpp.sms+xml".

The details of how the IRI-POI performs a target match when the target is not a non-local ID is described in TS 33.128 [4] clause 4.12.2.8.2.3.

The non-LI aspect of the flow diagrams shown in the subsequent clauses are based on the flow diagrams defined in 3GPP TS 24.341 [5].

4.2.5.1.3.2.2 Originating network without status report

The flow-diagram shown in figure 4.2.5.1.3.2.2-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message over IP to the Party_B (not shown) without a status report.

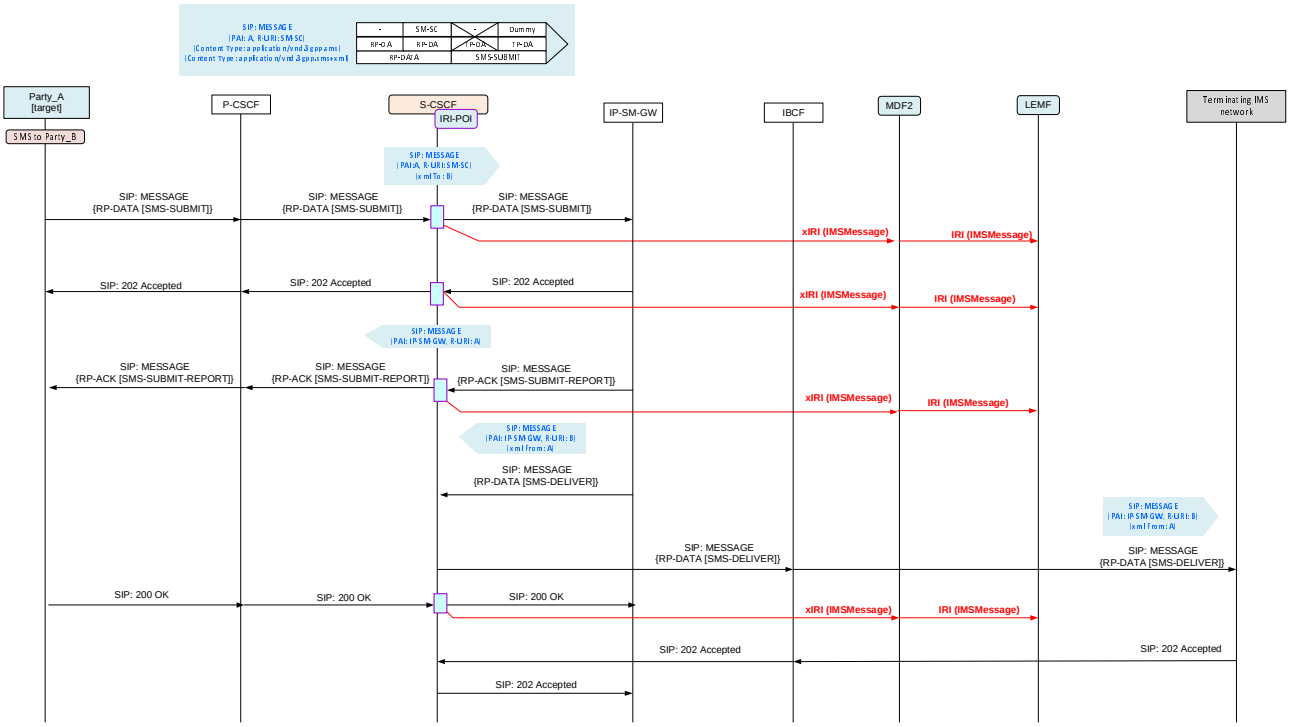


Figure 4.2.5.1.3.2.2-1: SMS over IP message transfer in originating network without a status report

The IRI messages in the above flow-diagram have the same correlation number.

The P-Asserted Identity and From fields of the SIP MESSAGE that includes the SMS-SUBMIT have the A party identity (that happens to be the target in this flow). The Request URI and the To fields of the SIP MESSAGE that includes the SMS-SUBMIT-REPORT have the A party identity (which happens to be the target in this flow).

The <To> field within the XML body of the content type "vnd.3gpp.sms+xml" includes the destination address which is mapped to the Request URI field by the IP-SM-GW of the SIP MESSAGE sent to the S-CSCF toward the destination.

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.3.2.2-2 illustrates the sequence of steps involved when Party_A sends an SMS message over IP to Party_B (not shown) who happens to be a target non-local ID without a status report.

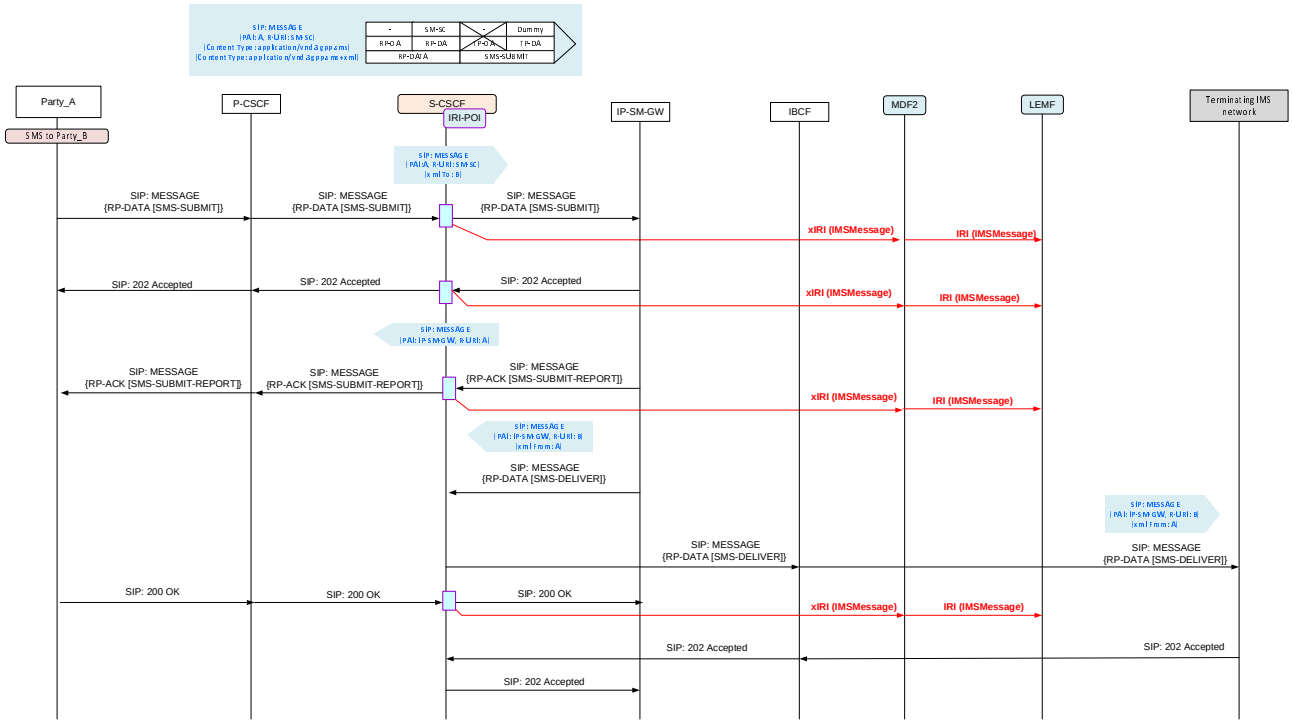


Figure 4.2.5.1.3.2.2-2: SMS over IP message transfer in originating network without a status report (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The <To> field within the XML body of the content type "vnd.3gpp.sms+xml" includes the destination address (that happens to the target non-local ID in this flow). The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-SUBMIT-REPORT later sent back the A-party is also intercepted.

4.2.5.1.3.2.3 Originating network with a status report

The flow-diagram shown in figure 4.2.5.1.3.2.3-1 illustrates the sequence of steps involved when Party_A (target) sends an SMS message over IP to the Party_B (not shown) with a status report.

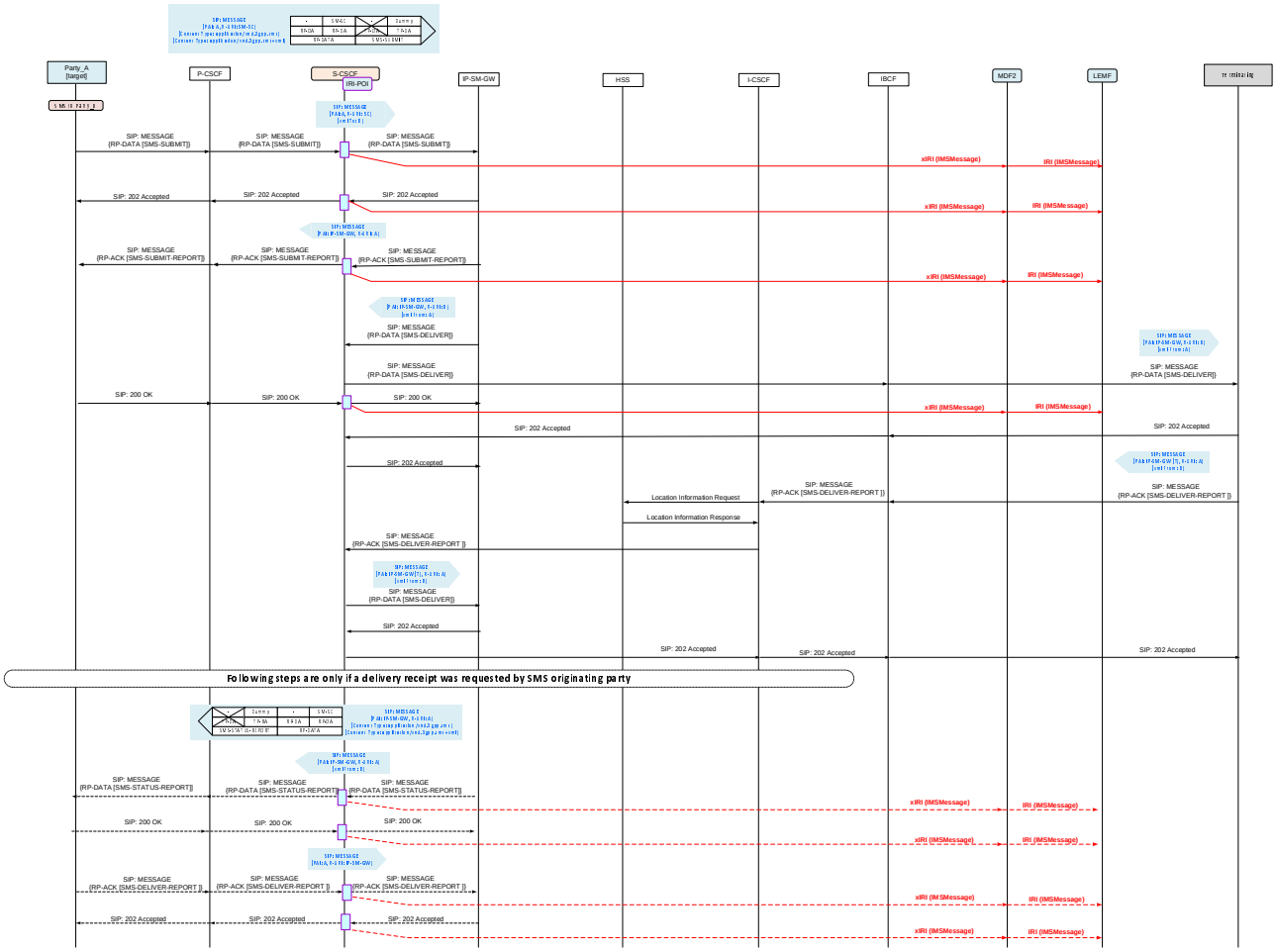


Figure 4.2.5.1.3.2.3-1: SMS over IP message transfer in originating network with a status report

The IRI messages in the above flow-diagram have the same correlation number.

The first part of this flow is same as the flow shown in figure 4.2.5.1.3.2.2-1. In the second part of the flow, the Request URI and the To fields of the SIP MESSAGE that includes the SMS-STATUS-REPORT have the A-party identity (that happens to be the target in this flow). The P-Asserted Identity and the From fields of SIP MESSAGE that includes the SMS-DELIVER -REPORT have the A party identity (which happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The flow-diagram shown in figure 4.2.5.1.3.2.3-2 illustrates the sequence of steps involved when Party_A sends an SMS message over IP to Party_B (not shown) who happens to be a target non-local ID with a status report.

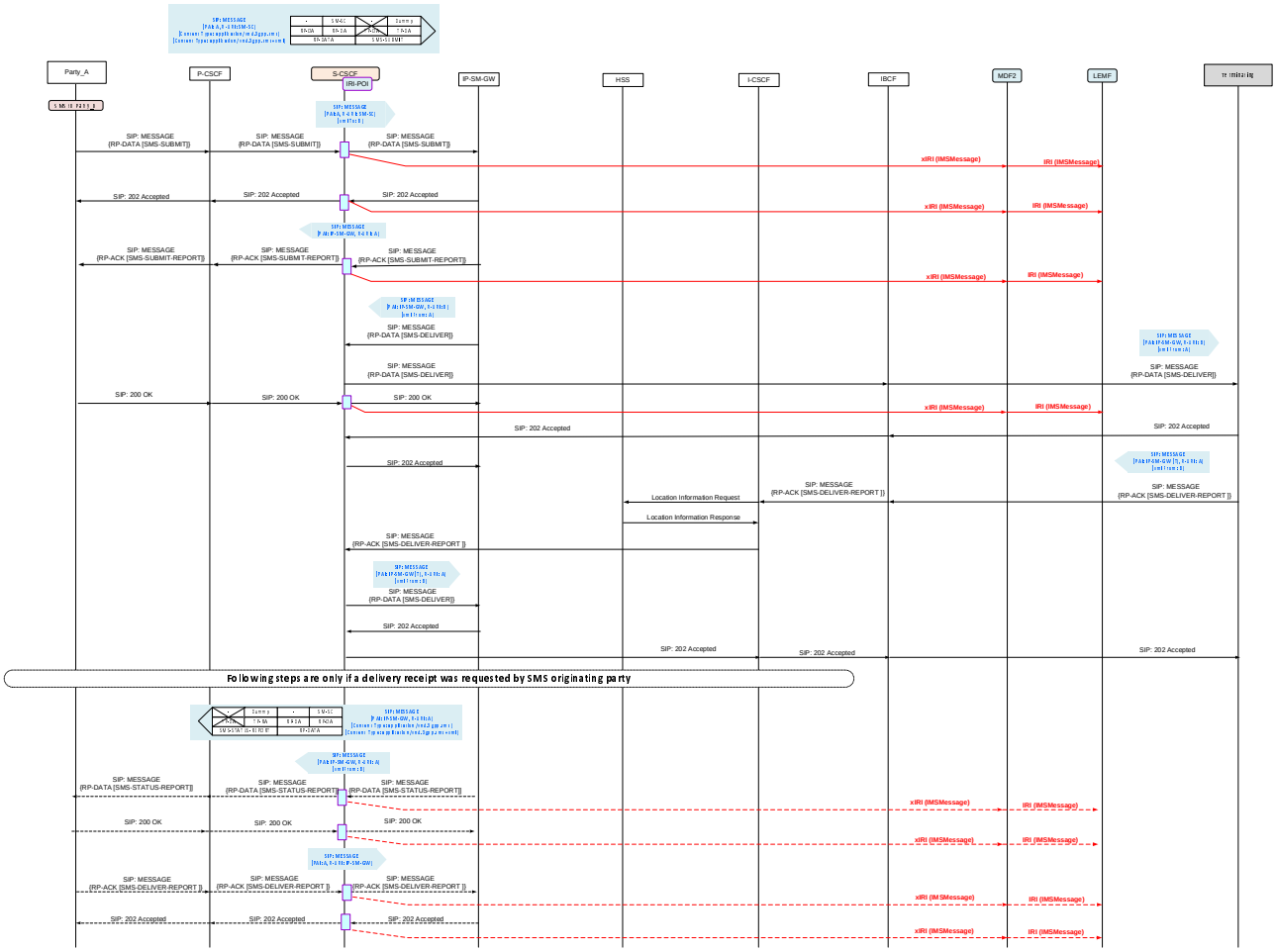


Figure 4.2.5.1.3.2.3-2: SMS over IP message transfer in originating network with a status report (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The first part of this flow is same as the flow shown in figure 4.2.5.1.3.2.3-1. In the second part, the <From> field within the XML body of the content type "vnd.3gpp.sms+xml" includes the destination address (that happens to the target non-local ID in this flow) of the associated SMS message sent by the A-party. The IRI-POI retains and uses this identity to determine that the SIP MESSAGE with the SMS-DELIVER-REPORT later sent back the SMS-GMSC is also intercepted.

4.2.5.1.3.2.4 Terminating network

The flow-diagram shown in figure 4.2.5.1.3.2.4-1 illustrates the sequence of steps involved when Party_B (target) receives an SMS message from Party_A (not shown).

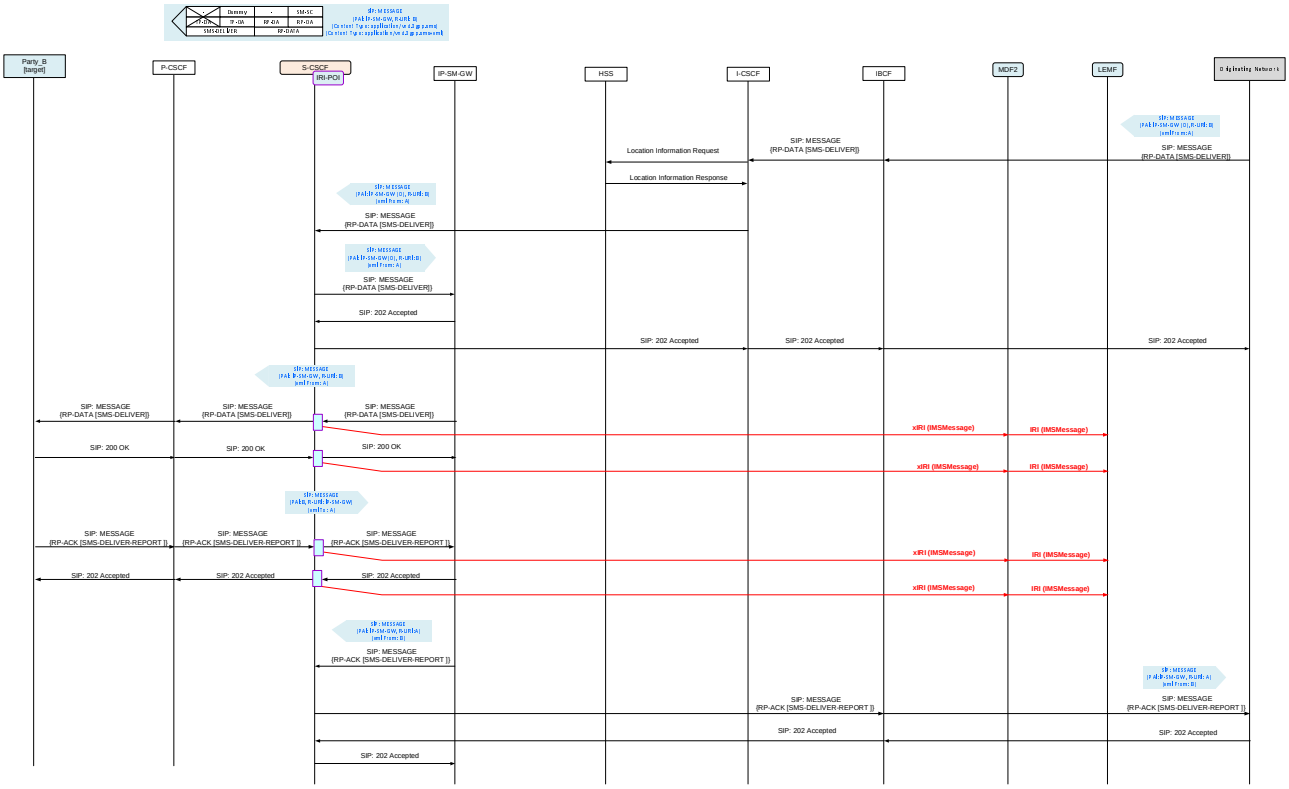


Figure 4.2.5.1.3.2.4-1: SMS over IP message transfer in terminating network

The IRI messages in the above flow-diagram have the same correlation number.

The Request URI and the To fields of the SIP MESSAGE that includes the SMS-DELIVER have the B party identity (that happens to be the target in this flow). The P-Asserted Identity and the From fields of the SIP MESSAGE that includes the SMS-DELIVER -REPORT have the B party identity (that happens to be the target in this flow).

NOTE: Not all the fields of SIP MESSAGE are shown in the above and below flow-diagrams.

The <From> field within the XML body of the content type "vnd.3gpp.sms+xml" includes the origination address which is mapped by the B-party to the <To> field within the XML body of the "vnd.3gpp.sms+xml" in the SIP MESSAGE for the SMS-DELIVER-REPORT.

The flow-diagram shown in figure 4.2.5.1.3.2.4-2 illustrates the sequence of steps involved when Party_B receives an SMS message from Party_A (not shown) who happens to be a target non-local ID.

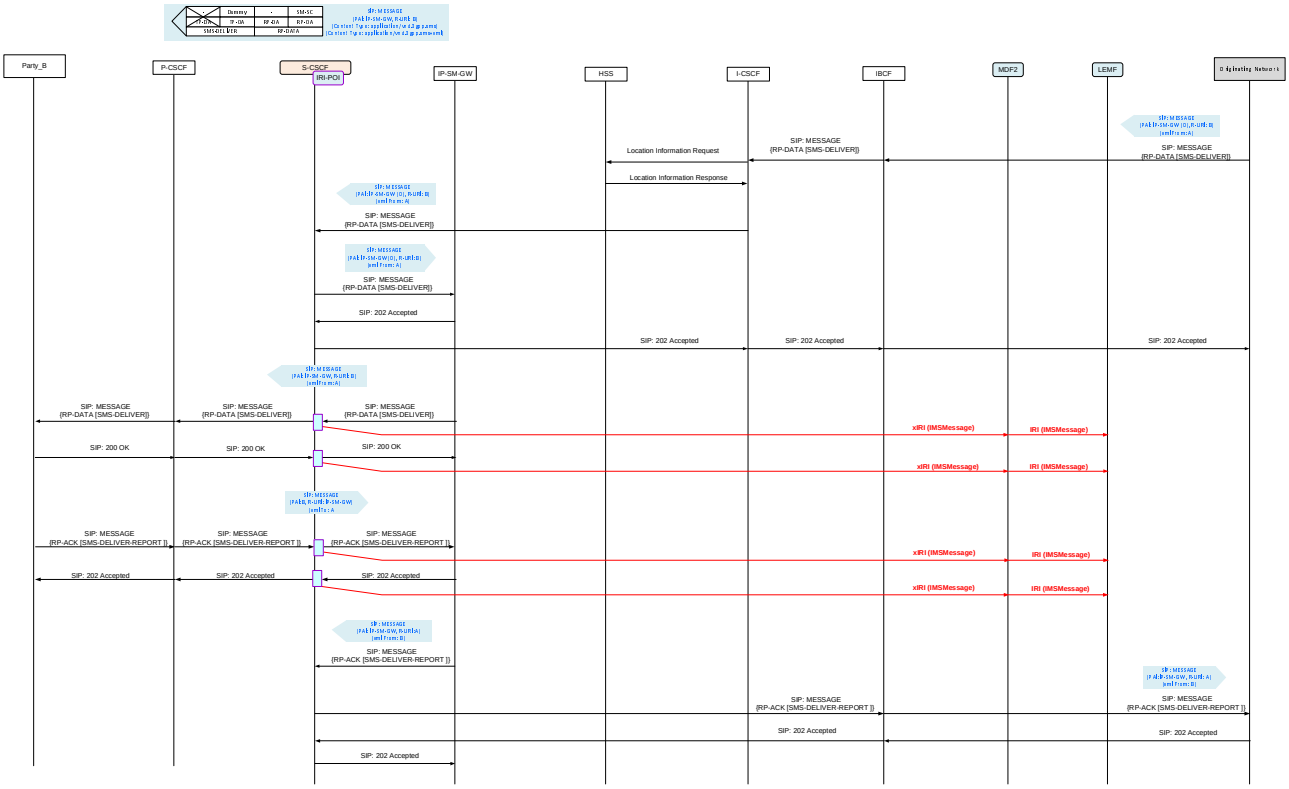


Figure 4.2.5.1.3.2.4-2: SMS over IP message transfer in terminating network (target non-local ID)

The IRI messages in the above flow-diagram have the same correlation number.

The <From> field within the XML body of the content type "vnd.3gpp.sms+xml" of the SIP MESSAGE for SMS-DELIVER includes the origination address (that happens to the target non-local ID in this flow). The <To> field within the XML body of the content type "vnd.3gpp.sms+xml" of the SIP MESSGE for SMS-DELIVER-REPORT includes the origination address (that happens to the target non-local ID in this flow).

4.2.5.1.3.2.5 Interception of SMS Command

As illustrated in clauses 4.2.5.1.3.2.2 to 4.2.5.1.3.2.4 the SMS message transfer does not involve the SM-SC when the destination is addressed using a SIP URI. When the SMS-SC is not involved, the SMS is not stored for a later delivery. Accordingly, SMS-COMMAND procedures do not apply to this case.

The TS 24.341 [5] allows the IP-SM-GW at the originating end to transfer the SMS over SM-SC in the event direct delivery to the destination fails. In that case, the SMS-COMMAND procedures do apply. The procedures in that case are same as illustrated in clause 4.2.5.1.2.2.5.

4.2.5.1.4 Instant messaging mapped to SMS

Figure 4.2.5.1.4-1 provides the generic architecture showing the major NFs involved in the transfer of SMS messages from the originating end to the terminating end. In this case, the originating end initiates an instant message which the IP_SM-GW (originating end) would route to the SM-SC (basically, the mapping the instant messaging to SMS).

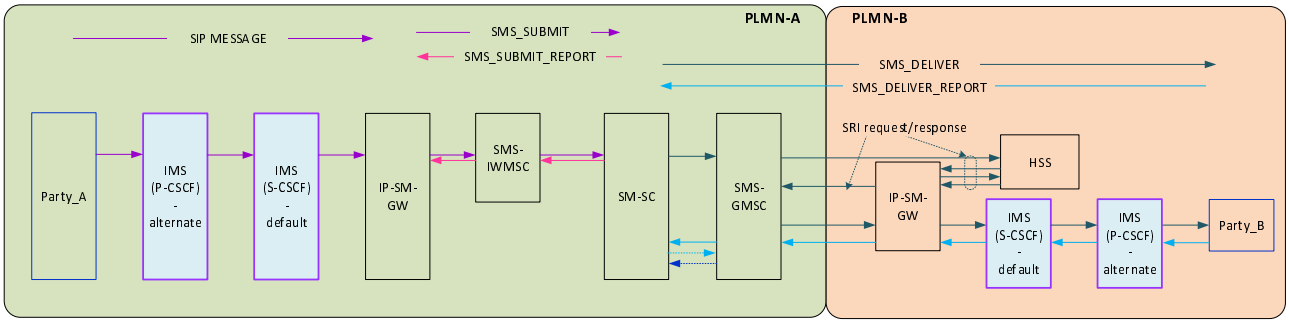


Figure 4.2.5.1.4-1: Generic architecture for instant messaging mapped to SMS

In the default deployment option (see TS 33.127 [3] and TS 33.128 [4]), the S-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in the PLMN-A or when Party_B is a target non-local ID in the PLMN-A. Likewise, the S-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in the PLMN-B or when Party_A is a target non-local ID in the PLMN-B.

In the alternate deployment option (see TS 33.127 [3] and TS 33.128 [4]), the P-CSCF in the originating network (PLMN-A) provides the IRI-POI functions when Party_A is a target in the PLMN-A or when Party_B is a target non-local ID in the PLMN-A. Likewise, the P-CSCF in the terminating network (PLMN-B) provides the IRI-POI functions when Party_B is a target in the PLMN-B or when Party_A is a target non-local ID in the PLMN-B.

As can be observed from the figures from 4.2.5.1.2.1-1 and 4.2.5.1.4-1, the NFs involved in the message transfer from the originating UE to the IP-SM-GW and from the IP-SM-GW to the terminating UE are the same for the two cases. Therefore, no separate diagrams are used to illustrate the roaming scenarios of SMS over IP with SIP URI as the destination. The illustrations shown in figures 4.2.5.1.2.1-2, and 4.2.5.1.2.1-3 may be used to understand the topology views when the UEs are roaming (Home-Routed or LBO).

Annex A:
Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2024-07	SA3LI#94	S3i240506				Initial version	0.0.1
2026-07	SA3LI#102	S3i260358				Agreed for Change Control	0.0.2
2026-08	SA#113	SP-260729				Submitted to SA#113 Plenary for Approval	1.0.0
2026-09	SA#113	-				Approved by SA#113 Plenary for Change Control	19.0.0

History

Version	Date	Status
V19.0.0	October 2026	Publication